Non-invasive Techniques for Monitoring Different Aspects of Sleep: A Comprehensive Review

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Quality sleep is very important for a healthy life. Nowadays, many people around the world are not getting enough sleep, which has negative impacts on their lifestyles. Studies are being conducted for sleep monitoring and better understanding sleep behaviors. The gold standard method for sleep analysis is polysomnography conducted in a clinical environment, but this method is both expensive and complex for long-term use. With the advancements in the field of sensors and the introduction of off-the-shelf technologies, unobtrusive solutions are becoming common as alternatives for in-home sleep monitoring. Various solutions have been proposed using both wearable and non-wearable methods, which are cheap and easy to use for in-home sleep monitoring. In this article, we present a comprehensive survey of the latest research works (2015 and after) conducted in various categories of sleep monitoring, including sleep stage classification, sleep posture recognition, sleep disorders detection, and vital signs monitoring. We review the latest research efforts using the non-invasive approach and cover both wearable and non-wearable methods. We discuss the design approaches and key attributes of the work presented and provide an extensive analysis based on ten key factors, with the goal to give a comprehensive overview of the recent developments and trends in all four categories of sleep monitoring. We also collect publicly available datasets for different categories of sleep monitoring. We finally discuss several open issues and future research directions in the area of sleep monitoring.

CCS Concepts: • **Human-centered computing** → *Ubiquitous and mobile devices;*

Additional Key Words and Phrases: Sleep monitoring, sleep stages, sleep disorders, sleep postures, vital signs

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1 INTRODUCTION

Sleep is a vital element in daily human life, as essential as water and food. Humans spend one-third of their life in sleep, as they need 6–8 h of sleep daily for a healthy lifestyle. Quality sleep is also essential. Low-quality or inadequate sleep leads to impaired physical health and may cause cognitive and psychological problems. People who do not get proper sleep are more prone to different chronic diseases such as obesity, diabetes, and hypertension [95]. Spiege et al. [83] showed that inadequate sleep could increase the risk of obesity and diabetes. Vorona et al. [90] demonstrated that there is a relation between sleep time and obesity. Brooks et al. [12] asserted that sleep apnea is a high-risk factor for hypertension. Inadequate sleep is also one reason for road accidents and lower productivity [39, 44]. Due to the importance of sleep, this area has become very popular among the research and development communities. Many studies have been conducted to analyze and understand the quality and behaviour of sleep. It has become a separate branch of medicine and has gained a lot of attention recently [34, 94].

There are various sub-topics in the area of sleep monitoring. Different studies have focused on one or more of these sub-topics. Some studies focus on sleep stages classification, and they are more interested in investigating the sleep cycles [1, 10]. Sleep cycles are fundamental, and any anomaly in sleep cycles or less time in a specific cycle can lead to disorders causing health problems. Some studies focus on monitoring different postures during sleep [18]. They investigate the body movements and postures such as supine, prone, left, and right and the time spent in each pose during sleep. Posture monitoring can provide useful information about sleep behaviour and is vital for preventing pressure ulcers and sleep apnea [56]. Other studies focus on the detection of sleep disorders [63]. These studies investigate sleep-related disorders such as **obstructive sleep apnea (OSA)**, insomnia, and restless leg syndrome. The detection of these disorders is very challenging as the patients are unaware of these disorders. One other group of studies focuses on monitoring vital signs such as respiration and heart rate during sleep [34]. Vital signs are correlated with sleep behavior and can provide important information about sleep quality and some disorders.

From the related literature, we can find that various techniques have been used to monitor sleep. The traditional approach to analyze and understand sleep quality is to use clinical methods [19]. Clinical methods apply specialized hardware and environment and are supervised by trained people, usually health workers. The patient has to stay overnight in the clinic. This approach is both complex and costly. Nowadays, smart health technologies offer ubiquitous solutions for sleep monitoring, which are relatively cheap and easy to use [64, 67]. Innovations in intelligent sensing have led to the development of many sensors that can be used as wearable or embedded in bed frames. These sensors can easily monitor sleep by capturing the related data. These solutions can be used for in-home sleep monitoring and do not require trained technicians.

In this article, we review and analyze non-invasive and in-home studies conducted for sleep monitoring. We focus on the techniques that do not require the users to wear different types of invasive sensors, which can cause disturbance in natural sleep or require special environment. We divide the related literature into four categories: *sleep stages*, *sleep postures*, *sleep disorders*, and *vital signs monitoring*. We review the state-of-the-art works in these fields and provide the reader an overview of the advancements in these areas of sleep monitoring. We discuss and compare both wearable and non-wearable (device-free) methods used for monitoring sleep and provide a comprehensive comparison of these techniques using ten key factors. We also identify the possible research gaps and provide future research directions for the research community. To ease the reading, a list of acronyms used throughout this article is given in Table 1.

The rest of the article is organized as follows. Section 2 discusses the related work, and Section 3 provides the background of sleep monitoring and the clinical approaches used for sleep monitoring. We also collect some publicly available data sets in this section. Section 4 presents the four categories of sleep monitoring in detail. We review the latest literature for both wearable and non-wearable solutions in this section. Section 5 discusses several future research directions, and Section 6 offers some concluding remarks.

Table 1. List of Acronyms and the Corresponding Definitions

Acronyms	Definition	Acronyms	Definition
PSG	Polysomnography	KNN	k-Nearest Neighbors
OSA	Obstructive Sleep Apnea	CRF	Conditional Random Field
IoT	Internet of Things	PSQI	Pittsburgh Sleep Quality Index
RFID	Radio Frequency Identification	NB	Naive Bayes
BCG	Ballistocardiography	BN	Bayesian Network
EEG	Electroencephalogram	RF	Random Forest
ECG	Electrocardiogram	DST	Dynamic State Transition
EOG	Electrocardiography	COTS	Commercial Off-the-shelf
EMG	Electromyography	PSD	Power Spectral Density
NSPR	National Sleep Research Resource	RSSI	Received Signal Strength Indicator
RF	Radio Frequency	FMCW	Frequency Modulated Continuous Wave Radar
UWB	Ultra Wide Band	HMM	Hidden Markov Model
CSI	Channel State Information	WHO	World Health Organization
RSS	Received Signal Strength	HR	Heart Rate
AASM	American Academy of SleepMedicine	RR/BR	Respiration Rate/Breathing Rate
REM	Rapid Eye Movement	PCA	Principal Component Analysis
NREM	Non-Rapid Eye Movement	mmWave	Millimeter Wave
PPG	Plethysmography	ICA	Independent Component Analysis
LSTM	Long Short-term Memory	FFT	Fast Fourier Transform
BLSTM	Bidirectional Long Short-term Memory	ACF	Auto-Correlation Function
IMU	Inertial Measurement Unit	ICSD	International Classification of Sleep Disorders
DT	Decision Tree	PLM	Periodic Limb Movements
SVM	Support Vector Machine	CMW	Characteristic Moment Waveform
RBF	Radical Basis Function	TCW	Time Characteristic Waveform
CNN	Convolutional Neural Network	RLS	Restless Leg Syndrome
RNN	Recurrent Neural Network	GBR	Gradient Boosting Regressor
DNN	Deep Neural Network	DWT	Discrete Wavelet Transform

2 RELATED WORK

There are several surveys that summarize the works done in sleep monitoring and focus on different aspects of sleep monitoring. Kelly et al. [45] presented a review article summarizing the recent developments for in-home sleep-monitoring devices. This study mainly focuses on the commercially available devices and categorizes them in brain signal-based, autonomic signal-based, movement-based, and bed-based systems. This work also presents the discussion about using other consumer devices that can be used for sleep monitoring. The authors highlight research gaps in the sleep monitoring area and concluded that portable devices are increasing for sleep monitoring. Still, these devices and systems need to be standardized. Surantha et al. [85] conducted a survey on the **Internet of Things (IoT)**-related works for sleep monitoring. The emphasis of this work is on the quality of sleep. The authors discussed a few sleep-related disorders and the clinical techniques for sleep monitoring. This work presents an IoT architecture for sleep quality monitoring. Ibanez et al. [36] focused on sleep questionnaires and diaries. This survey only reviews the self-reporting approaches for sleep assessment and provides some critical analysis of these techniques' accuracy and validation.

Matar et al. [57] presented a comprehensive review of the unobtrusive methods used for sleep monitoring. This survey addresses three main topics: sleep stages, sleep disorders, and monitoring techniques. The authors categorized the related literature based on physiological signals captured during sleep, such as breathing, cardiac signals, and body movements, and review the various studies in each of these categories. The authors also discussed the challenges and limitations of the techniques presented for each category. Sadek et al. [74] presented a survey on the use of non-intrusive technologies for sleep monitoring. The article discusses both wearable (e.g.,

Table 2. Summary of the Previous Surveys and This Survey

Surveys	Year	Main Focus	Data sets	Comparisons	Sleep Stages	Sleep Disorders	Vital Signs	Postures	Clinical Methods	In-home Methods	
										Wearable	Non-wearable
[45]	2012	Consumer devices	No	No	Yes	No	No	No	No	Yes	Yes
[85]	2016	Sleep quality	No	No	No	Yes	No	No	Yes	No	No
[36]	2018	Questionnaires and diaries	No	Yes	No	Yes	No	No	No	No	No
[57]	2018	Physiological signals (actigraphy)	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
[74]	2019	Consumer devices	No	No	No	No	No	No	No	Yes	Yes
[19]	2019	Computational analysis methods	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
[88]	2019	Sleep apnea	No	No	No	Yes	No	No	No	No	Yes
[3]	2020	Sleep apnea	No	Yes	No	Yes	Yes	No	Yes	Yes	Yes
[62]	2020	Wireless body area network	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Our Survey	2021	Ubiquitous methods	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

bracelets) and non-wearable (e.g., bed-embedded) approaches but the main focus of this work is consumer devices and various gadgets available in the market for sleep assessment. There is very less content on the research aspect of the discussed technologies. Fallmann et al. [19] proposed a survey on sleep monitoring techniques. This survey discusses sleep behaviour in terms of movements, stable state, and disorders. The authors categorize the literature into clinical and home-based approaches and summarize different techniques from each category. However, the main focus of this survey is computational analysis methods for sleep assessment. The authors discussed various computational studies presented for sleep stages classification, body movements monitoring, and disorders detection and list the challenges and limitation of those techniques.

Tran et al. [88] presented a survey on the Doppler radar-based monitoring techniques for the detection of sleep apnea. This article describes the architecture of radar-based system and discusses the different aspects of the system such as noise removal and signal processing. The authors categorized the radar-based techniques into four categories based on their methodology. These categories are time-frequency analysis, which uses time-series and frequency domain in the signal processing, numerical analysis, which uses numerical techniques, classification, and training, which uses machine learning and other methodologies, which use experimental and mathematical modeling. Ahmadzadeh et al. [3] conducted a survey on biomedical sensors, technologies and algorithms for the breath-related sleeping disorders. The article categorizes the literature into invasive and non-invasive techniques and provides a review of the related articles in both the categories. The authors also discuss the different challenges involved in the sensors selection, data processing, and classification algorithm used in the presented literature. Pan et al. [62] presented a review of the current techniques and future challenges in the area of sleep monitoring. The authors reviewed the related literature on sleep stages classification and sleep disorders and address some of the challenges faced by the community in the current systems for sleep monitoring. The article also discusses clinical and wearable sensor-based approaches for sleep monitoring and proposed their own system for sleep monitoring. The proposed system consists of various wearable sensors attached to both wrists, both legs and chest position and uses machine learning-based approach to monitor sleep stages, body movements, and snoring.

There is a growing interest in the area of sleep monitoring and several surveys have been conducted recently. Table 2 gives a summary of these surveys. As can be seen from the table, these surveys focus on different aspects of sleep monitoring like disorders detection, sleep stages classification or posture recognition during the sleep. Also, the techniques reviewed by these surveys are different and focus on one or more aspects of sleep monitoring. None of these surveys discussed the datasets for sleep monitoring, which is a very important part of research in this field. To the best of our knowledge, there is no survey paper that provides a complete review of the work conducted in all the four sub-categories of sleep monitoring using the non-invasive approach. Due to this reason,

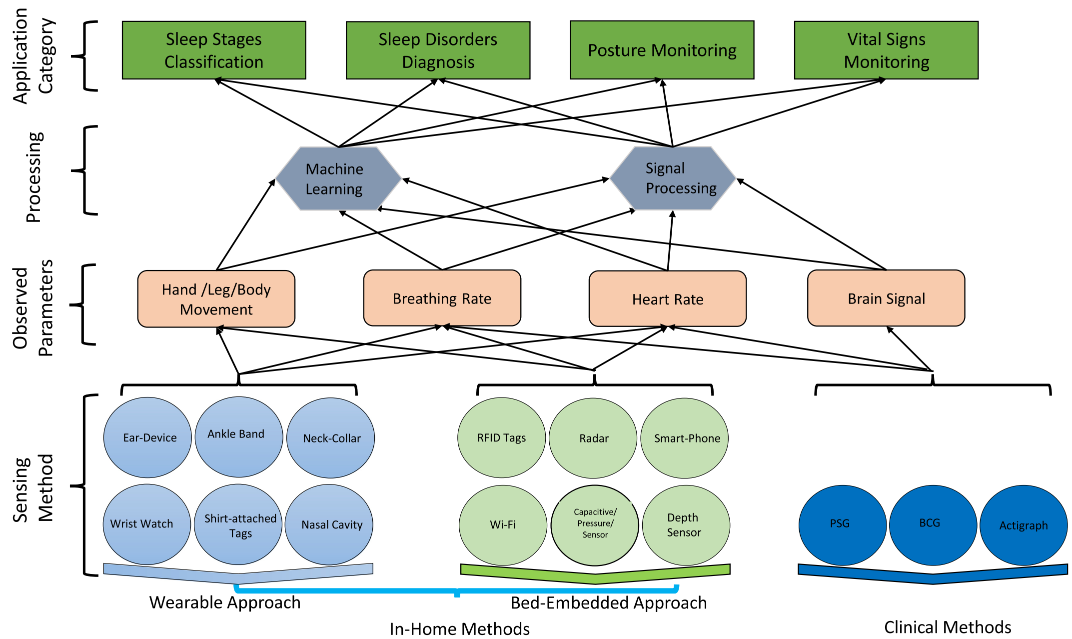

Fig. 1. General architecture of sleep monitoring system.

we believe that there is a need for a comprehensive review to give readers an overview of the latest research efforts in all the sub-categories of sleep monitoring, especially using the unobtrusive techniques. In this article, we review the recent works in sleep monitoring in the four sub-categories of sleep monitoring, which are sleep stages classification, sleep disorders detection, posture recognition, and vital signs monitoring.

3 BACKGROUND

In this section, we provide the background of sleep monitoring. We also discuss the clinical methods used for sleep monitoring. Due to the well-recognized importance of quality sleep and high prevalence of inadequate sleep, sleep monitoring has become a very important area of research. Many studies have been conducted that proposed different techniques for monitoring different aspects of sleep.

The general architecture of a sleep monitoring system can be divided into four layers as shown in Figure 1, consisting of (1) sensing, (2) observed parameters, (3) processing, and (4) application. The first layer deals with the choice of sensing method. Different sensors have been used in the literature to capture the information about the sleeping. In clinical methods, various sensors are used in an invasive method to capture the physiological signal while the subject is asleep. In-home methods use both wearable and bed-embedded devices. In the wearable method, different sensors are worn by the subjects on their body at different positions such as wrist, leg, ankle, chest and neck. The sensors (e.g., pressure, capacitive, Radio Frequency Identification (RFID) tags can be embedded in the bed frame or the sleeping mattress as well. The second layer of the architecture deals with the type of parameters observed. The sleeping involves various factors such as body movements, hand movements, leg movements, breath rate, heart rate and brain signals. Based on the objective of the system, one or more of these parameters can be captured. The third layer consists of the processing of the data captured. Two main approaches are used in the literature for processing the data. These approaches are machine learning and signal processing. The final layer consists of application category in which the final output of the system is decided. The output can be one or more of the given four categories.

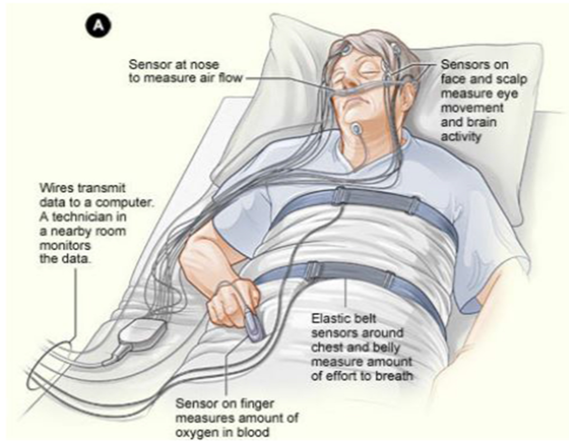

Fig. 2. Polysomnography.[1]

3.1 Clinical Approaches for Sleep Monitoring

Clinical methods refer to those methods that are conducted in special environments (labs/clinics/hospitals) under the supervision of trained staff. Currently, these methods have the highest accuracy and are considered as standards for sleep monitoring. These methods can be termed as invasive, because they involve various sensors attached to different body parts of the subject (e.g., head, nose, fingers, toes, and chest). These sensors cause disturbance in the natural sleep and cannot be used for long term monitoring. In this section, we discuss some of the important clinical methods used for sleep monitoring.

3.1.1 Polysomnography. The current gold standard technique used for sleep analysis and diagnosis of sleep disorders is **Polysomnography (PSG)** [77]. PSG is a complex clinical process that requires the subject to sleep for hours wearing many sensors as shown in Figure 2. This procedure can be conducted in specialized environment under the supervision of trained people (sleep experts and technicians). The patient has to stay overnight and the data from **electroencephalogram (EEG)**, **electrocardiography (EOG)**, **electromyography (EMG)**, nasal airflow, and pulse oximetry is recorded via the attached sensors. The data is then processed to analyze the sleep behaviour. PSG is not an ideal procedure for sleep analysis as the subject has to wear many different kinds of sensors and has to stay overnight in the specialized environment. This procedure is time-consuming, complex, and expensive and cannot be used for long-term [22].

3.1.2 Actigraphy. Actigraphy is another approach that has gained popularity in health care solutions [82]. Actigraphy is widely accepted as standard in medical domain for objective sleep quality measurement [19]. In this approach, a device that mainly consists of an accelerometer (can also use other inertial sensors such as gyroscope or magnetometer) is attached to the body (preferably wrist or ankle position) of the person and can record all the physical movements of the person. The normal assessment consists of seven consecutive days to get a representative image of the user's sleep. The data is recorded and then processed using machine learning or signal processing techniques for sleep analysis. This method has the advantage that it can monitor the person's sleep for longer period without any extra cost or effort. However, it can only provide information about the wake and sleep activity labels and the physical movements during the sleep [78].

[1]Source: https://www.sleep-apnea-guide.com/polysomnogram.html.

Table 3. Summary of the Publicly Available Data Sets

Data set	Year	Focus	Sensor Used	Deployment	# Subjects	Sample Size	Age range
[69]	1997	OSA, Sleep stages	PSG	Specialized environment	6,000	NA	40
[26]	2000	General purpose	PSG	Specialized clinic	16	8–10 h per subject	32–56
[66]	2000	Apnea	ECG	Specialized clinic	32	34, 313 min	NA
[30]	2011	Apnea, sleep stages	PSG (Jaeger-Toennies system)	Specialized clinic	25	5.9 to 7.7 h per subject	28–68
[61]	2014	General purpose	PSG	Specialized clinic	200	One full night per subject	18–76
[11]	2014	General purpose	Accelerometer	Wrist-worn	42	45 nights	28–86
[87]	2016	Sleep stages	Bioradar	Bedside	32	NA	22–67
[46]	2016	General purpose	PSG	Specialized clinic	100 + 8 + 8	1, 2, and 1 session for each subject	NA
[68]	2017	Sleep postures	Pressure sensor (Vista Medical FSA SoftFlex 2048)	Sleeping mat under mattress	13	23,400	19–34
[108]	2017	Sleep stages	Radio	Room	255	90,000 (30-s epoch)	NA
[24]	2018	Arousal	PSG	Specialized clinic	994	1,983	55 (mean)
[31]	2018	Respiration	RF (UWB-IR,WiFi CSI, Zigbee RSS, and sub-1 dB quantized RSS)	Either sides of the bed	20	160 h	55–60

3.1.3 Ballistocardiography. **Ballistocardiography (BCG)** is another clinical technique that is based on the signal generated by the heart when it pushes the blood into the vessels. This is a very old technique but recently got significant importance and has been used for vital signs and sleep monitoring [25]. This approach provides non-invasive monitoring of cardiac signals and does not require the users to wear any sensors on their bodies. Different sensors are employed to capture BCG signals such as fiber brag grating sensors, electromechanical film sensors, piezoelectric film sensors, microbend fiber optic sensors, polyvinylidene-fluoride sensors, hydraulic and pneumatic sensors, and pressure sensors [38, 49]. These sensors can be embedded in ambient environment such as bed and chairs. BCG method is mostly used for monitoring the cardiac activity and can provide good results but it involves a lot of sensors to be deployed, which makes it costly. Also, these sensors can be affected by different factors such as bed frames, mattress thickness, body movements, bed-partners, and motion artifacts.

3.2 Publicly Available Data Sets for Sleep Monitoring

Nowadays, there are many data sets publicly available in various research areas. These data sets are playing a vital role in the advancement of research as the research community can use these data sets to evaluate their newly proposed solutions. Like other fields of science, there are publicly available data sets for sleep monitoring as well. Most of these data sets consist of PSG recordings and are collected in specialized environment (hospitals/clinics). The National Sleep Research Resource (NSRR)[2] offers free access to many sleep related data-sets. Table 3 provides details about some of the publicly available data sets for sleep monitoring.

4 SLEEP MONITORING

There are four categories of sleep monitoring (see Figure 1). In this section, we review the latest work conducted in these categories.

[2]Source: https://sleepdata.org/datasets.

Table 4. Five Stages of Sleep

Stage		Description
Wake	W	The person is awake but on the bed
NREM	N1	Dosing off, which lasts for 1 to 5 min
	N2	A more subdued state in which the body is relaxed. The body temperature drops and the breathing, heart rate, and brain activity starts to slow down. N2 can last for 10–25 min
	N3	Deep sleep. Muscles are relaxed and breathing and brain activity slow down. N3 can last for 20–40 min and is the most important part of the sleep for different bodily processes.
Rapid Eye Movement (REM)	REM	In this stage, the whole body is paralyzed except the eyes and the brain. Dreams occur at this stage. REM is responsible for around 25% of the total sleep in adults.

4.1 Sleep Stages Monitoring

A very important aspect in sleep monitoring is the classification of sleep stages, which can help in the analysis of the sleep behaviour and diagnosis of sleep disorders. Initially, the sleep was divided into seven stages by the R-K method proposed by Rechtschaffen and Kales in 1968 [70]. Later in 2007, the **American Academy of Sleep Medicine (AASM)** [37] updated their manual for sleep scoring and divided the sleep into five stages. These stages are wake, **rapid eye movement (REM)** and three stages (N1, N2, and N3) of **non-rapid eye movement (NREM)** as shown in Table 4. Sleep cycle in healthy adults last for around 90 to 100 min, and it starts with N1 stage, which is the shortest, followed by N2 (which lasts for around 10 to 25 min) and then N3 stage. After N3, the REM stage counts for 20–25% of the total sleep. Monitoring and identification of these sleep stages is very critical for diagnosis of different sleep disorders and it can also help in tracking the treatment's response in patients [108]. In this section, we review the work conducted for sleep stages classification using the ubiquitous approach.

4.1.1 Wearable-based Approaches for Sleep Stage Classification. Due to the advancements in the production of wearable devices, the use of wearables has been increased significantly. Many applications, especially human activity recognition [35, 54] and healthcare [2], are now adopting the wearable approach because of the low-cost and non-complex operation. In this section, we review the wearable-based techniques used for sleep stage classification.

Beattie et al. [7] conducted a study for sleep stage estimation using the optical **pulse plethysmography (PPG)** and accelerometer. They collected the overnight sleep data from 60 adults who wore the devices (Fitbit) on both of their wrists. The authors use the Embletta for validation (ground truth) purposes, which collects the EEG, ECG, and EOG signals. Two sleep experts were hired to score the PSG records from the Embletta device as per the AASM guidelines. The labeling from the experts is used as the ground truth for the valuation of the proposed approach. This work uses three sets of features which are from movement, heart rate and respiration. These feature are fed into a linear discriminant classifier which can predict one of the four classes (Wake, REM, Light, and Deep) with 69% accuracy. One of the main findings of this study is that actigraphy can provide some information about the sleep stages (such as awake and sleep) but cannot accurately predict all the stages. By adding the physiological data (heart rate and breath rate) captured by PPG, the sleep stages can be detected with reasonable accuracy. Zhang et al. [106] proposed a technique for sleep stage classification based on actigraphy and heart rate captured by a wearable device. To get the ground truth, five sleep experts analyzed the PSG data and assigned the five sleep stages independently. The low-level features in both time and frequency domain are extracted from accelerometer and heart rate data, which are then combined to learn the mid-level features. A **bidirectional long short-term memory (BLSTM)** architecture is trained on the final features to classify the sleep data into one of the five stages. The authors evaluate the performance of the proposed system for two different groups; the resting group went to sleep without doing any exercise while the comprehensive group exercised before going to bed. They achieved an F1 score of 64% for the resting group and 60.5% for the comprehensive group.

Boe et al. [9] presented a technique based on wearable wireless senors to detect different stages of the sleep. This work uses **inertial measurement unit (IMU)** devices attached to different positions on the body and collects the physiological data including the body temperature during the sleep. PSG is used to get the ground truth. A population-based bagging classifier with **decision tree (DT)** is used to classify the sleep data into different classes. This work has evaluated the proposed approach for two-stage (wake versus sleep), thee-stage (wake versus NREM versus REM) and four-stages classification (wake versus light versus deep versus REM) and the performance was compared with that of Actiwatch. The results show that their approach outperforms the Actiwatch but the performance of the proposed technique is poor for the four-stage classification. Zhang et al. [107] designed their own wearable multi-sensor vest called SensEcho for sleep stage classification. The SensEcho uses two electrodes attached to the chest and abdomen region and one oximeter attached at wrist position. An accelerometer is also embedded in the vest. This multi-sensor vest captures the ECG and the breathing signal. A number of different features including both time and frequency domain features are extracted form the raw data. The extracted features are fed into a BLSTM, which can assign one of the four stages with 80.75% accuracy. The authors also evaluated the proposed approach on a publicly available data set and achieved the same accuracy. Nakamura et al. [59] proposed another wearable technique called Hearables, which uses an in-ear sensor for sleep stages classification. This in-ear device has two electrodes and can easily record the ECG signal for the over-night sleep. After the pre-processing, different features are extracted, which includes structural complexity features and spectral features. The final features are fed into a **support vector machine (SVM)** with **radial basis function (RBF)** kernel, which classify the sleeping data into five classes with 74% accuracy. PSG is used as a ground truth.

4.1.2 Non-wearable-based Approaches for Sleep Stage Classification. A major issue with the wearable approach is that users are bound to wear the device all the time. Nowadays, many solutions are using the device-free (non-wearable) approach [35], in which the users are not required to wear anything. Non-wearable devices-based approaches have the advantage of having the least disturbance in natural sleeping, because the users are not required to wear any sensor. The sensors devices are either placed in the environment (room) or embedded in the bed frame or mattress. Sleep monitoring can be done via a single device such as radar and smart phone or multiple sensors can be integrated in the bed. The most commonly used sensors that are used as non-wearable for sleep monitoring include pressure sensor, radar, Wi-Fi, radio, load cells, force sensors, smart phones, and so on. In this section, we discuss the non-wearable-based techniques used for sleep stage classifications.

Zhao et al. [108] proposed a radio signal-based technique for predicting the sleep stages without attaching anything to the subject body. This technique uses a radio device placed in the bedroom while the user is sleeping and then analyzes the reflected signals to extract the information about the sleep. One of the main contribution of this work is the feature engineering. The authors use game theory and adversarial training for discarding all the irrelevant information in the reflected signal and extract only the relevant features. This work uses a combination of **Convolutional Neural Network-Recurrent Neural Network (CNN-RNN)** in their model architecture and achieves an accuracy of 79.80%. Zhang et al. [105] presented a non-contact technique for sleep stage classification using the Doppler radar. A continuous-wave Doppler radar is placed near the user bed, which emits a single tone signal. The reflected signal contains the information about the breathing of the user, because the chest movements induce disturbances in the signal. The reflected signal also captures the movements of the user. Different features related to body movements, respiration rate, and heart beat are extracted from the reflected signal, which are fed into a bagged tree classifier. The classifier can assign one of the four classes with 78.6% accuracy.

Yi et al. [100] proposed the use of a hydraulic sensors for monitoring the sleep stages in a non-invasive way. The hydraulic sensors are embedded in the bed under the mattress, which can capture the body movements as well as the movements caused by breathing and BCG. The captured data is first cleaned by passing through Butterworth filter to remove the low and high frequency noise. Different features including both time and frequency domain are calculated for respiration and heart rate. Two classifiers, SVM and **k-nearest neighbors (KNN)**, are used in both single layer and hierarchical configurations to predict the three stages of sleep. Single layer SVM achieved the highest accuracy of 85.5%. A similar approach is presented by Gargees et al. [23], which also uses the hydraulic sensors embedded in the bed frame to classify the different sleep stages. After cleaning the BCG data captured by the hydraulic sensors, it is passed through a deep model architecture, which consists of CNNs and **long short-term memory (LSTMs)**. A pre-trained CCN (trained on the sleep data of 56 subjects from another study) is employed using the transfer learning technique. The study shows that transfer learning can boost the performance of the classifier and can achieve an accuracy of 90.80%. Zhang et al. [104] presented the design and implementation of a system called SMARS, which uses ambient radio signals for sleep stage classification. The basic working principle of this system is the tiny changes induced in the reflected signal due to breathing and body movements. This work employs a statistical model that accounts for all the reflecting and scattering multi-paths to accurately estimate the breath rate and hence the sleep stages. Awake and sleep stages are detected based on the frequency of movement, i.e., frequent movements means the person is awake. For the classification of REM and NREM stages, an SVM with RBF kernel is used, which can predict the two classes with 88% accuracy.

One other group of researchers exploited the use of inertial sensors embedded in the smartphone for sleep monitoring [14, 15, 27, 28]. Gu et al. presented a phone-based system called SleepHunter [27], which can monitor different stages of sleep. Using the embedded sensors in the smartphone, SleepHunter captures the specific body movements and acoustic signals (breathing) and combines them with environmental factors to detect different sleep stages such as REM, light sleep, and deep sleep). This work uses a statistical model called a **conditional random field (CRF)** to depict the relationship between sleep events (body movement, acoustic, illumination, time, and personal factor) and the sleep stages. The experiments performed show that the given model can achieve an accuracy of 64% for detecting different sleep stages. Chang et al. [14] also presented a smartphone-based work called iSleep for sleep quality monitoring. This work is the extension of their previous work [28] and uses the built-in microphone in the smartphone to capture the acoustic signal during sleep. iSleep first cleans the captured data by removing the ambient noise and then extracts different statistical features from the cleaned acoustic signal. These features are fed into an event detection system that employs a decision tree classifier and can detect different sleep-related events such as coughing, snoring, and body movements. The proposed model uses these events to score the quality of life for both short-term (single nights) and long-term using two sleep scoring criteria: **Pittsburgh Sleep Quality Index (PSQI)** and actigraphy. iSleep can be used for monitoring single users as well as two users sleeping together. The authors have performed various experiments to evaluate the performance of iSleep.

Summary. A summary of the work presented for sleep stage classification is given in Table 5 (wearable-based) and Table 6 (non-wearable-based). From the presented work, we can see that solutions with higher level of obtrusiveness and cost have the better accuracy. Also, these systems can recognize more stages than solutions with low level of obtrusiveness. This can be explained by the level of physiological signal captured by these complex systems. Another observation is the increasing use of machine learning and feature engineering in this area. More solutions are now focusing on the feature engineering aspect and then use the existing models. Accelerometer is the widely used sensor in wearable solutions while non-wearable solutions are mostly using the pressure sensors. Although, the wearable systems are easy to use, the accuracy is low. More work is required to increase the accuracy of these solutions to an acceptable level.

Table 5. Comparison of the Wearable-based Solutions for Sleep Stage Classification

Work	# Stages	Sensor-Placement	Device Used	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
[7] 2017	4	Accelerometer, PPG-wrist	Fitbit	Linear discriminant classifier	69%	Low	Low	Movement, breathing, heart rate	60	PSG by Embletta MPR
[106] 2018	5	Accelerometer, PPG- wrist	Microsoft Band I	BLSTM-based RNN	60.50%	Low	Low	Movement, heart rate	39	PSG by Compumedics.
[9] 2019	2,3,4	IMU, Temperature-chest, wrist, ankles and forehead	BioStampRC, Thermochron iButton, ActiWatch Spectrum (for comparison)	Bagging classifier with DT	60.60%	Medium	Medium	Movement, tempera-ture,ECG	11	PSG
SenseEcho [107] 2019	4	Accelerometer, ECG-chest, oximeter-wrist	SenseEcho	BLSTM-based RNN	80.75%	Medium	Medium	Breathing, ECG	32	PSG
Hearables [59] 2020	5	EEG-ear	Self-made	SVM with RBF kernel	74.10%	Low	Medium	ECG	22	PSG by SOMNO screen

Table 6. Comparison of the Non-wearable-based Solutions for Sleep Stage Classification

Work	# Stages	Sensor-Placement	Device Used	Contact	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
[27] Slee-pHunter 2015	3	Microphone, accelerometer, light sensor, clock-besides pillow	Samsung S4,hfill Samsung Note 2, HTC G14, MIUI2SC	No	CRF, SVM	64.5%	Low	Low	body movement, breathing, illumination	45	Zeo
[108] 2017	4	Radio-room	COTS	No	Game theory, CCN, RNN, adversarial training	79.80%	Low	Low	Breathing, heartbeat from reflected signal	25	EEG-based sleep monitor
[105] 2017	4	Radar-room	Digital-IF Doppler radar	No	Bagged tree	78.60%	Low	Low	Movement, respiration, heartbeat from reflected signal	1	PSG
[100] 2019	3	Pressure sensor-bed	Hydraulic bed system	Yes	SVM with cubic kernel, KNN	85%	Low	Medium	Respiration, heartbeat	5	PSG
[23] 2019	3	BCG, bed	Hydraulic bed system	Yes	Transferhfill learning, CNN,hfill LSTM, DNN	90.80%	Low	Medium	Respiration, heart rate	5	PSG
[14] iSleep 2020	3 events	Microphone-besides pillow	Nexus smartphones	No	Signalhfill processing, DT	90%	Low	Low	acoustic signal	7	Ipod touch, Snowball USB microphone
SMARS [104] 2021	3	Radio-room	Atheros WiFi chipsets	No	SVM with RBF, Statistical model	88%	Low	Low	Movement, respiration from reflected signal	6	PSG, EMFIT, ResMed

4.2 Sleep Posture Monitoring

Recognizing sleep postures is a very important factor in the assessment of sleep. It provides very useful information about the quality of sleep. Posture recognition is of particular importance for bedridden patients to avoid pressure ulcers also called pressure sores. Prolonged stay for elder people and post-surgery patients in the bed can cause skin problems due to laying in one posture for longer time. Monitoring the postures can help the care givers to provide better support and avoid the pressure ulcers and other skin problems. Due to its importance, many studies have been conducted to monitor the sleeping postures. In this section, we review some of the recent works done for posture recognition.

4.2.1 Wearable-based Approaches for Posture Monitoring. In this section, we discuss the wearable-based techniques used for sleep posture monitoring. Sun et al. [84] presented a sleep monitoring system called SleepMonitor by using the accelerometer embedded in a smartwatch. The working of this technique is based on the idea that wrist position is correlated with the body postures. SleepMonitor uses the wrist positions to find the sleep postures. Different statistical features are extracted from the accelerometer data, which are used to train four different classifiers including **Naive Bayes (NB)**, **Bayesian network (BN)**, DT, and **random forest (RF)**. The RF performs better than the other models and recognizes the four sleep postures with 91.70% accuracy. Fallmann et al. [20] also used the accelerometer-based approach for sleep posture recognition but this work attaches the senors to both ankles and the chest. This work uses **Matrix Learning Vector Quantization (GMLVQ)** model, which is a distance-based classifier, to recognize one of the eight classes.

SleepGuard [13] also uses an IMU embedded in the smartwatch to monitor different sleep related activities including the postures. SleepGuard uses the arm position to recognize different sleep postures. In addition to recognize the four basic sleep postures, SleepGuard can also recognize three hand positions. Moreover, the proposed system can monitor other important sleep factors such as body rollover, micro movements, acoustic events (snoring, coughing, etc.), and illumination condition. The work conducted by Jeon et al. [41] uses wearable devices attached to both wrists and the chest. The wearable device consists of IMU sensors. A two-level classifier based on **Dynamic State Transition (DST)**-framework is used to classify the sleep postures and motions. First, the data is classified into **Sleep Posture Change (SPC)** and NonSPC groups using Random Under Sampling Boost (RUSBoost) algorithm. In the second level, SVM and RF algorithms are used to further classify SPC into eight and NonSPC into four motions. iSleePost [40] is another non-invasive system that uses inertial sensors attached to the wrist and chest positions of the users. The chest-attached sensor is only used in the training phase and is no longer needed after the model is trained. Basic features are extracted form the accelerometer data and two machine learning models (SVN and RF) are used to detect the four sleeping postures.

4.2.2 Non-wearable-based Approaches for Posture Monitoring. In this section, we present the non-wearable-based techniques used for sleep posture monitoring. Xu et al. [96] presented a matching-based approach called **Body-Earth Mover's Distance (BEMD)** for sleep posture recognition using the pressure image captured by the pressure sensors embedded in the bed sheet. BEMD treats the pressure image as 2D weighted image and calculates three descriptors (planar, polar and projection) before applying the histogram normalization. Euclidean distance and EMD are combined to evaluate the similarity of sleep postures. Finally, a skew-based posture classifier is used to classify the postures in one of the six postures. The skew-based classifier consists of **k-nearest neighbour (KNN)** and skew rate classifier. Barsocchi et al. [6] presented an unobtrusive system for posture recognition using the **force sensing resistors (FSR)** embedded in the bed frame. FSR captures the pressure caused by the human body (when lying in the bed). A multi-class logistic regression model is trained to classify the pressure data into one of the four postures. Liu et al. [53] presented a sleep monitoring system using WiFi signal. This work exploits the **channel state information (CSI)** collected through a **commercial off-the-shelf (COTS)** available WiFi device to detect different sleep postures and vital signs. The proposed system has two main modules: posture detection and vital signs estimation. For posture detection, the various statistical features are selected from the CSI using the Fisher score method. The number of features is further reduced using the principal component analysis method, which selects only the most effective features. Various machine learning models such as discriminant analysis, KNN, SVM, and random forest are trained to detect one of the four postures (curl-up, supine, prone, and recumbent). For RR estimation, the signal is first cleaned by applying various filters, and then a threshold-based method is used to extract the sub-carriers most sensitive to the breathing motion. The RR is estimated using a modified peak detection method, which filters out the fake peaks. For two persons, the RR is estimated using K-means clustering (k = 2) with **power spectral density (PSD)** and corresponding frequency as 2-D features. The HR is calculated from the high-frequency component of the CSI using different filters and PSD.

Hu et al. [33] proposed an RFID-based system to recognize sleep postures. This work attaches multiple (8×8) passive RFID tags to the blanket, which can capture the posture information of the sleeping person. When the person moves or change sides in the bed, the **received signal strength indicator (RSSI)** of the received signal

Table 7. Comparison of the Wearable-based Solutions for Sleep Posture Recognition

Work	# Posture	Sensor-Placement	Device Used	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
SleepMonitor [84] 2017	4	Accelerometer-wrist	Sony Smartwatch 3, Huawei Watch	NB, BN, DT, RF	91.70%	Low	Low	Wrist position	16	Smartwatch attached at belly, Comparison with BioWatch
[20] 2017	8	Accelerometer-legs, chest	Shimmer3	Generalized Matrix Learning Vector Quantization	98%	Medium	Low	Body position	6	Video camera
SleepGuard [13] 2018	4+3	Accelerometer, gyroscope, orientation sensor-wrist	Self-made	Euclidean distance, template matching	90%	Low	Low	Arm position	15	Video camera
Sleepcare Kit [42] 2019	4+8	Accelerometer, gyroscope, magnetometer-both wrists and chest	AHRS EBIMU24GV	Directional features, two-level classification, Random Under Sampling Boost, SVM, RF	95%	Medium	Low	Body position	5+11	Kinect
iSleePost [40] 2021	4	Accelerometer-wrist, chest	Koala	SVM, RF	85%	Medium	Low	Body position	1	Accelerometer

at the reader will be disturb accordingly. This work exploits the changes caused in the received signal as a result of the posture changes and apply CNN after removing the noise. The CNN can divide RSSI image (each pixel corresponds to a tag value) into one of the five postures with 86% accuracy. Liu et al. [52] proposed a similar approach called TagSheet in which they embed multiple passive RFID tags into the sleeping mat/bed sheet in the form of a matrix. The system takes grey-scale snapshots of the variance in the RF signal from all the tags, which can represent the different postures. This work employs various image processing techniques such as Gaussian blur, Ostu-based binary conversion of the grey-scale image, and the removal of scattered pixels. TagSheet then applies multi-level hierarchical recognition by first classifying the postures into three coarse-grained classes and then refine it by further classifying to get the total six classes. One of the advantage of this system is that no prior data or training is required and the system can be used in a plug and play manner.

Yue et al. [103] conducted a study exploring the wireless signals and proposed a system called BodyCompass for sleep posture monitoring in a home environment. BodyCompass uses a **frequency-modulated continuous wave radar (FMCW)** radio, which transmits a low-power signal. These signals are reflected from the environment as well as the person body. BodyCompass analyzes these reflected signals by using different signal processing and machine learning techniques to infer the sleep postures. This work calculates the sleep postures in terms of angle, i.e., the angle between the bed surface and the trunk of the sleeping person. If the angle is 0, then the posture is facing upward, while 90 angle means posture is facing rightwards. BodyCompass first extracts filtered multi-path features specific to a person from the RF reflection. A fully connected neural network is trained on the data of a single user in a specific home. Then transfer learning is used to train the model for predicting the posture of new users. The use of transfer learning reduces the requirement of training data for new users. BodyCompass is evaluated in various settings and achieves an accuracy of around 94%.

Summary. Tables 7 and 8 provide a summary of the work presented for sleep postures monitoring. Like other categories of sleep monitoring, the wearable-based posture monitoring solutions are mostly using the accelerometer while the non-wearable-based solutions are using pressure sensor and RF technology. Different solutions are able to recognize different postures ranging from 4 to 12 postures but almost all solutions presented can recognize the four basic postures with high accuracy. Validation is one concern for posture recognition system as there is no benchmark data set available and most solutions either have to use a video camera to record the ground truth or use another device to get the ground truth. Video cameras have the privacy concern while other devices may not give the 100% accurate results. Machine learning plays an important role in posture recognition as well. Data can be collected through various sensors but it is the machine learning that can recognize the

Table 8. Comparison of the Non-wearable-based Solutions for Sleep Posture Recognition

Work	# Posture	Sensor-Placement	Device Used	Contact	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
BEMD [96] 2016	6	Pressure sensor-bedsheet	Piezo-electrical pressure sensors	Yes	EMD, Euclidean distance, KNN, Skew rate classifier	91.21%	Low	Medium	Pressure image	14	NA
[6] 2016	4	Force sensing resistors-bed	Self-made	Yes	Logistic regression	91.20%	Low	Medium	Pressure values	3	Video
[53] 2018	4	WiFi-room	COTS WiFi	No	Signal processing, sub-carrier selection, PCA, KNN, peak detection	98%	Low	Low	CSI	6	Observation
[33] 2018	4	RFID tags-blanket	COTS	Yes	CNN	86.24%	Low	Low	Breathing from the reflected signal	1	NA
Tagsheet [52] 2019	6	RFID tags-bedsheet	Impinj H47	Yes	Image processing, polynomial fitting, hierarchical classification, PCA	96.70%	Low	Low	Breathing from the reflected signal	12	Video
BodyCompass [103] 2020	NA	Radio-room	No	FMCW radio	Filtered multipath extraction, majority voting, fully Connected NN	94%	Low	Low	Breathing from the reflected signal	26	Accelerometer

postures. Some of the commonly used machine learning algorithms in posture recognition are KNN, RF, SVM, NB, and **hidden Markov model (HMM)**.

4.3 Vital Signs Monitoring

Monitoring vital signs such as **respiration rate (RR)** and **heart rate (HR)** during sleep is very important. These signs provide information about the overall health of the person and can be used as indicators for the diagnosis of many disorders. RR can help in the diagnosis of different sleep disorders. Different people have different HR, depending on person's health and age. An abnormal HR may indicate the possibility of some kind of health issues such as heart problem. According to the **World Health Organization (WHO)**, around 17 million people die every year because of cardiovascular disease, which counts for 31% of all the deaths occurring each year. Monitoring HR is very important, because it can provide crucial information about the health of the person and can help in diagnosing heart problem at early stages. Also, these vital signs can provide information about different sleep stages as they change in different stages. Owing to the importance of vital signs during sleep, many works have been conducted in recent days to monitor the vital signs. In this section, we review some of the latest works done for vital signs monitoring using the ubiquitous approach.

4.3.1 Wearable-based Approaches for Vital Signs Monitoring. This section presents the wearable-based works for vital signs monitoring during the sleep. Sharma and Kan [80] proposed an RF-based technique for vital signs monitoring using a single passive RFID tag attached to the shirt of the user at the chest area. This work uses the **near-field coherent sensing (NCS)** concept for extracting the HR and RR without requiring the skin contact. Different filters are applied to extract the HR and RR from the phase values of the reflected signals. Melici et al. [58] presented a magnetometer-based system that can monitor the breathing rate by measuring the variations in magnetic field. The magnetometer is packaged in a wearable device along with an RFduino and is worn around the body using an elastic chest belt. After applying different filters, a peak detection algorithm is used to calculate the respiration rate. The performance of the proposed system is compared with a temperature airflow sensor and the results indicate that the proposed system can monitor the RR with reasonable accuracy. RF-ECG [92] is another technique that uses passive RFID tags attached to the shirt (at chest area) of the user for estimating the heart rate variability. The reflected signal from the passive tags is cleaned by removing the affect of breathing and the signals from multiple tags are fused together to strengthen the heartbeat signal. After applying other filtering techniques such as wavelet-based de-noising, a **principal component analysis (PCA)**-based template mechanism is applied to estimate the inter-beat intervals. Hussain et al. [34] proposed an

RFID-based technique for sleep monitoring. Two passive RFID tags are attached to user's shirt and the reflected signals from both the tags are captured by the reader. After removing the ambient noise, the reflected signals from both tags are fused together to strengthen the breathing signal. This work uses the **automatic multiscale-based peak detection (AMPD)** algorithm to estimate the breathing rate. The results of the experiments show that the proposed technique can achieve good accuracy for single as well as multiple users.

Chu et al. [16] presented a wearable sensor-based solution for monitoring respiration rate and volume. This work uses a disposable strain sensor [65], which is placed on the abdomen and ribs area and can capture the strain in the ribcage and abdomen area caused due to breathing (inhaling and exhaling). The proposed system can monitor the respiration rate and volume in various ambulatory conditions but requires prior calibration for every subject. Kiaghadi et al. [47, 48] presented a wearable solution called Phyjama, for physiological sensing using fiber-enhanced pyjamas. This work embeds pressure sensors (resistive patch) and triboelectric sensors (which are sensitive to very low pressure) at different parts of the clothing (pyjama), which are usually under pressure (pressed). Phyjama first finds the posture of the subject using a **decision tree (DT)** classifier, which can easily detect the posture by exploiting the difference of pressure exerted on each patch. The RR is calculated from each patch by counting the zero crossings in the cleaned signal. A median is taken from all the patches to find the final value for RR. For identifying a valid BCG peak, different features are calculated from the amplified signal, which includes sparse coding feature weights, envelope-based features for resistive patches and triboelectric sensors, posture information, and peak amplitude. These features are fed into an SVM classifier. The results from all the patches are fused together to find a more accurate result.

TagBreath [93] is another work that uses passive RFID tags for breath rate monitoring. TagBreath uses three tags attached to the user's shirt for capturing the chest motion resulting from breathing. Unlike Reference [34], which uses the RSSI values, TagBreath uses the phase values from the reflected signal and fuses the signals from all three tags. After applying multiple filters in both frequency and time domain, the breath rate is estimated by calculating the zero crossing in the selected window.

4.3.2 Non-wearable-based Approaches for Vital Signs Monitoring. In this section, we present the non-wearable-based studies for vital signs monitoring during the sleep. Jia et al. [43] presented a system called VitalMon for monitoring the vital signs using geophones, which can sense the vibrations in the bed caused by ballistic force. A distinguishing feature of the proposed system is that it can monitor the vital signs (HR and RR) of users even if they are sharing the bed with another user. VitalMon first separates the heartbeat signal (the carrier signal is the heartbeat, because respiratory signal's frequency is too low for the geophone sensing capability) for reach user using the DUET [73, 101] algorithm and then extracts the respiratory signal using the square-law demodulation approach. The authors implemented the proposed model and performed different real-world experiments to demonstrate the effectiveness of the model. SleepSense [51] is a non-contact sleep monitoring system that can monitor the breathing rate along with other factors such as bed exit and on-bed movements. This work uses a Doppler radar placed above the bed, which emits a single-tone carrier signal and uses the phase shifts in the reflected signal as a results of the chest movements due to respiration, to estimate the breathing rate. The phase shift is proportional to the movement displacement, which can be calculated by demodulating the phase using the an extended **differential and cross multiple (DACM)** algorithm. Different statistic, time, and frequency domain features are extracted and fed into a decision tree classifier, which detects one of the three classes: breathing section, bed exit, or on-bed movement. An adaptive-threshold-based peak detection algorithm is used to calculate the breathing rate in the breathing section.

Yang et al. [99] presented mmVital, which is a contact-less system for vital signs monitoring using the **millimeter wave (mmWave)**. mmVital uses the RSS of the reflected signals from the human body, which can capture the chest motion due to breathing. The proposed model analyses these reflected signals to estimate the vital signs (RR and HR) as well as the posture of the person. mmVital consists of two main modules: human finder and monitoring. The first task is to locate the human in the environment so that the signal can be directed in that direction. mmVital achieves this by profiling the indoor environment using the omni-sweep procedure. Once the human is located, the next step is to analyze the reflected signals from the human body, which contains information about vital signs. mmVital employs multiple filters to clean the signal and uses the peak detection method for estimating RR and HR. mmVital can also estimate the four postures (front, right, left, back) using a decision tree-based classifier. The apnea is detected by analyzing the amplitude of the breathing signal as the signal will be flat in the case of central apnea and with low amplitude in the case of hypopnea. Yue et al. [102] proposed an RF-based solution called DeepBreath for respiration monitoring. The proposed system is capable of estimating the respiration rate of multiple subjects even if they are sitting very close to each other. DeepBreath uses an FMCW radio equipped with multiple antennas, which transmits a wireless signal and captures its reflection from the users in the observation area. The proposed system consists of three main modules. The first module is the breath separation in which the breathing signals of each users is separated using the **independent component analysis (ICA)**. The second module deals with motion detection, because breathing signals are affected when the users move (e.g., change sides). A convolutional neural network is trained to detect the movements of the target users while ignoring the irrelevant (surroundings) movements. The last module maps the extracted breathing signals to the corresponding subjects (as there are multiple subjects). DeepBreath achieves this by mapping it as an optimization problem and solves it using the dynamic programming. Yang et al. [98] proposed a technique for respiration monitoring using COTS WiFi devices. This work can estimate the breathing rates for multiple persons simultaneously. The key focus of this work is the optimal deployment of the WiFi transceivers such that each person is at good location of only one transceiver and bad location for the other transceiver resulting in the simultaneous monitoring of multiple persons each by a separate transceiver.

The work conducted by Sadek and Mohktari [76] uses fiber optic embedded in a sensor mat to monitor different sleeping factors including RR and HR. The proposed system uses the mean and standard deviation to identify the sleep movement and empty bed states. In the sleep state, a Chebyshev band-pass filter is used to extract RR and HR signals. The HR is calculated using the Maximal overlap discrete wavelet transform while the RR is calculated using the peak detection algorithm after smoothing the signal using different filters.

Alizadeh et al. [4] presented a mmWave-based system for remote monitoring of vital signs.This work uses an FMCW radar that emits a directed beam and gets reflected from human body capturing the minor chest motion caused by breathing. After passing the reflected signal through various filters (for removing distortions), an advanced phase unwrapping procedure is applied to extract the vital signs (RR, HR). Helena [17] is a recent work that uses geophone (seismic sensor) integrated in the bed frame to monitor sleep activities including the RR and HR. Helena first determines the bed status and if the user is on the bed and there is no movement, the RR and HR are then calculated using an envelop-based estimation method. After obtaining the envelop, HR can be calculated using the peak detection algorithm. Since the seismometer is insensitive to lower frequency measurements (in case of respiration), RR cannot be directly estimated from the seismic data but instead the detected HR peaks are used to generate the respiration modulation signal with the help of extrapolation technique. Then the RR can be estimated by counting the peaks from the obtained envelop of the modulated signal. The authors have done extensive experiments to evaluate the performance of Helena. Experiments have been conducted in lab environments as well as in the wild settings and the results show that Helena can monitor the vitals signs with high accuracy and in real time. Turppa et al. [89] uses a FMCW radar for vital sign monitoring during sleep. This work uses **Fast Fourier Transform (FFT)**-based cepstral analysis method for extracting the HR. For the RR, an **auto-correlation function (ACF)** is applied on the phase signal of a selected range. The proposed system is evaluated in different scenarios resembling real-world situations including normal and abnormal cases.

Summary. A summary of the papers reviewed for vital signs monitoring is given in Tables 9 and 10. This category of sleep monitoring is dominated by wearable-based approach. Wearable solutions have high accuracy with low cost. RF technology is widely used in the wearable solutions in which passive RFID tags are attached

Table 9. Comparison of the Wearable-based Solutions for Vital Signs Monitoring

Work	Vital Signs	Sensor-Placement	Device Used	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
[80] 2018	HR,RR	RFID-chest	NA	Signal processing	97.58%	Low	Low	RSSI	1	Observation
[58] 2018	RR	Magnetometer-chest	Self-made	Signal processing	85%	Medium	Low	Magnetic vectors	1	Air-flow sensor
RF-ECG [92] 2018	HRV	RFID tags-shirt	ImpinJ tags and reader	Signal processing, DWT, fusion, PCA-based template estimation	93%	Low	Low	Phase values	15	Heal Force PC-80A ECG Monitor
[34] 2019	RR	RFID tags-shirt	Alien tags and reader	Signal processing	95%	Low	Low	RSSI	4	Observation
[16] 2019	RR	Strain sensor-abdomen	self-made	Signal processing	NA	Medium	Low	abdomen strain	8	metronome, spirometer
Phyjama [48] 2020	HR, RR	pressure, triboelectric-clothes	self-made	Signal processing, sparse coding, fusion, DT, SVM	2bpm for HR, 1bpm for RR (median error)	Medium	Medium	Pressure	21 + 3	ECG, PPG
TagBreathe [93] 2020	RR	RFID tags-shirt	Impinj reader, Alien tags	Signal processing	95%	Low	Low	Phase values	4	Metronome mobile application

Table 10. Comparison of the Non-wearable-based Solutions for Vital Signs Monitoring

Work	Vital Signs	Sensor-Placement	Device Used	Contact	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
[43] VitalMon 2017	HR, RR	Geophone-bed-frame	SM-24 geophone	Yes	Signal processing, DUET algorithm, square-law demodulation	2.62 bpm for RR, 1.90 bpm for HR	Low	Low	vibration caused by heart beat	86	observation, pulse oximeter
SleepSense [51] 2017	RR	Doppler Radar-above bed	Self-made	No	Decision tree	93.45%	Low	Low	Phase shift in the reflected signal	3	Nasal airflow sensor
mmVital [99] 2017	HR, RR	mmWave-bedsides	Vubiq transmitter and receiver	No	Signal processing, peak detection, omni-sweep, decision tree	mean estimation error of 0.5bpm for RR and 2.5 bpm for HR	Low	Low	Reflected signal	7	pulse oximeter, Neulog respiration belt
DeepBreath [102] 2018	RR	radio-bedside	COTS FMCW	No	Signal processing, ICA, CNN, dynamic programming	0.139 median breathing rate error	Low	Low	Reflected signal	13 couples	SleepView
[98] 2018	RR	WiFi-bedside	5300 NIC, WR841N router	No	Gaussian movement model, Fresnel zone, hampel and wavelet filters, peak detection	1 bpm(MAE)	Low	Low	Changes caused in received signals by chest motion	5	Chest strap
[76] 2018	RR,HR	Optical fiber-sleeping mat	Self-made	Yes	Signal processing, maximal overlap discrete wavelet transform, peak detection	NA	Low	Low	pressure exerted by body	3	Self-reported survey
[4] 2019	HR, RR	Radar- above bed	Texas Instrument mm-wave radar	No	Signal processing, phase unwrapping	94% for RR, 80% for HR	Low	Low	Reflected signal phase	1	Hexoskin sesnor
Helena [17] 2020	RR,HR	Seismic sensor-bedframe	Self-made	Yes	Envelope-based HR/RR estimation, peak detection	1.26 bpm, 0.52 rpm- 2.41 bpm, NA (MAE),.	Low	Low	Pressure exerted by body	10+25+2	pulse oximeter (Zorvo), metronome, Apple watch
[89] 2020	RR,HR	FMCW radar-above the bed	Self-made	No	Signal processing, peak detection, hamming window, auto-correlation function, Hann window	3.6% for HR,9.1% for RR (MAE)	Low	Low	Changes caused in received signals by chest motion	10	PSG by Embla Titanium and VTT's BCG installed under the bedsheet

to chest or shirt of the users. Non-wearable solutions are also using the RF technology in the form of radars and radio devices installed in the room to capture the sleep data. Unlike the other categories of sleep monitoring, which mostly use machine learning, signal processing is used for vital signs estimation. One of the main focus of the presented works is the use of novel methods for removing the noise from the captured data, because the vital signs can easily be suppressed by the environment noise. Various filtering techniques have been presented to clean the signal and extract the RR and HR of the sleeping person. One concern here is the amount of data collected for the evaluation purposes. Most of the experiments presented in these works are simulating the sleep and not the actual sleep, i.e., the users are resting/sitting to simulate the sleep. There is a need of long experiments that involve the overnight sleep of the user to reflect real-world scenarios.

4.4 Sleep Disorders

Sleep disorders are different diseases that are attributed to lack of quality sleep and one of the main reasons of inadequate sleep problem. Sleep disorders also contribute to other diseases. Some of the most common symptoms of sleep disorders are irritability, depression or anxiety, tiredness, and daytime sleepiness [79]. According to the **International Classification of Sleep Disorders (ICSD-3)**, there are seven types of sleep disorders, which are sleep-related breathing disorders, insomnia, sleep-related movement disorders, circadian rhythm sleep-wakefulness disorders, central hypersomnolence disorders, parasomnia and other disorders [62]. These disorders can be cured with proper treatment but the diagnosis of these disorders is a challenging task. Recently, many in-home techniques have been proposed that can provide comparable results with clinical methods. In this section, we review some of the latest works conducted for detection of sleep disorders using the unobtrusive approach.

4.4.1 Wearable-based Approaches for Sleep Disorder Detection. In this section, we review the wearable-based techniques used for sleep disorders detection. Kye et al. [50] presented a wearable band-based system to detect **periodic limb movements (PLMs)**. This band consists of inertial sensors (accelerometer and gyroscope) and is worn on the dorsum of the foot. This work proposes the **motion synchronized windowing (MSW)** technique in the feature extraction phase in which the data stream is segmented into intervals where the motion occurs. Different classifier are trained and tested but KNN achieves the highest accuracy for detection of PLMs in the performed experiments. AutoTag [97] uses the passive RFID tags attached to the chest area of the user for detection of sleep apnea. This work proposes a novel technique for mitigating the frequency hopping offset of RFID system. Unlike the other approaches, AutoTag uses an unsupervised recurrent autoencoder method for apnea detection and treats the breathing signal as a time sequence in a time window and uses an LSTM for encoding the breathing signal. The output of the LSTM is used to estimate the mean and variance vector, which in turn are used for computing the latent vector. Another LSTM is used for decoding the mean and variance vectors to get the reconstructed breathing signal. **Kullback-Leibler (KL)** divergence method is used to detect apnea by comparing the difference between original and reconstructed signals.

Fang et al. [22] proposed an apnea detection system based on the **characteristic moment waveform (CMW)** method. This work uses a microphone attached near the nose of the user to capture the acoustic signals when the user breaths. In the prepossessing stage, the amplitude contrast of the original signal is reduced by using different signal processing techniques and then CMW is used for RR estimation based on **time characteristic waveform (TCW)**. Ren et al. [71, 72] presented a smartphone-based work for fine-grained sleep monitoring. This work uses smartphone earphones that are placed closer to the users while sleeping. These earphones can capture the breathing sound (acoustic signal) of the users, which are then processed in the smartphone to detect breathing rate and other sleep events (e.g., snore, cough, apnea). The captured signal is cleaned and the environmental noise is removed by applying multiple filters. The breathing rate is estimated by using the envelope detection method. For detecting sleep events, various acoustic features are extracted and used to train an SVM model, which can predict the presence of the sleep event in the captured signal. The authors evaluate the proposed system in three settings: wearing earphones, putting the earphone beside the pillow, and putting the earphones on the bedside. The results demonstrate that the system can achieve good results in all the settings with the best performance in the wearing earphones setup. Jeon and Kang [42] developed a wearable sleepcare kit, which can detect sleep apnea in real-time. This kit consists of two parts: a sensing device called Bio-Cradle and a control device called **personal activity assisting and reminding (PAAR)** band. The Bio-cradle consists of PPG and accelerometer sensors and a nasal tube for collecting physiological data while the PAAR works as a scheduler. Sleepcare kit is a plug and play system, which does not require any prior training data and can monitor the sleep in real time using a mobile application. This system uses the breathing and SpO2 level to detect apnea and hypopnea. The proposed system can also identify the level of apnea and can alert the caregivers in real-time using the mobile application.

Bobovych et al. [8] also developed a wearable device that can be worn on ankle position and can capture two types of leg movements: dorsiflexion (which are correlated with **restless leg syndrome (RLS)** and **Attention**

Deficit Hyper Activity Disorder (ADHD)) and complex leg movements (which are correlated with sleep fragmentation and brief arousal). Since the proposed system was supposed to be resource constraint (wearable), a single feature (standard deviation) is used to train an SVM classifier. The main focus of this work is not the processing part but rather the power usage analysis of the hardware and software.

4.4.2 Non-wearable-based Approaches for Sleep Disorder Detection. This section presents the non-wearable-based solutions for sleep disorders detection. Nandakumar et al. [60] presented a contact-less solution for apnea detection using the built-in speakers in the smartphones. The speaker in the phone is used as a sonar by emitting a frequency-modulated sound signal and the reflection is captured and analyzed to extract the breathing information of the user sleeping near by. A modified peak detection algorithm is used to detect different sleeping events such as central apnea, obstructive apnea and hypopnea based on the breathing pattern. Waltisberg et al. [91] presented a framework for the detection of sleep apnea and PLMS using the in-bed sensor system, which consists of strain gauges to detect the pressure changes caused by the breathing motion. This work explores both the measurement fusion and the decision fusion approaches. PCA and a two-layered feature extraction approach is used while minimum **redundancy maximum relevance criterion (mRMR)** filter is used for feature selection. An NB classifier is used to classify the sleeping event. The results show that both measurement and decision fusion approaches give same results and are better as compared to a rule-based approach using the standard deviation.

EZ-Sleep [32] uses an ambient radio device installed in the bedroom to monitor insomnia and other sleep activities. The radio device emits RF signals, which are reflected from the human body and the environment. EZ-Sleep analyzes these reflected signal to extract the sleep-related information such as bed location, bed-entry, bed-exit, and other insomnia assessment parameters. The bed is located first using a 2D histogram and image segmentation technique. Once the bed is located, the next step is to identify whether the user is lying on the bed or not. EZ-Sleep achieves this by using HMM. Before estimating the sleep parameters (vital signs, etc.), EZ-Sleep first needs to detect when the user fall asleep (sleep versus awake classification). This is done by using CNN with 14-layer residual network model and to further increase the accuracy for detecting onset sleep, **Gradient Boosting Regressor (GBR)** is build on the top of CNN. Results show that EZ-Sleep can detect different sleep activities with high accuracy. Swangarom et al. [86] proposed a contact-less system for insomnia detection using the fabric-pressure sensors embedded in the sleeping mat. The sensors can capture the body movements resulting from breathing and changing positions. The **Athens insomnia scale (AIS)** questionnaire is used for validation of the results. An SVM with polynomial kernel function is used for classifying the subjects into two classes; healthy and suffering from insomnia. Sadek et al. [75] proposed another non-invasive solution for apnea detection using the fiber optic sensors embedded in a mat and put under the mattress. This work uses the BCG principle and captures the cardiac movements (HR) and chest movements (RR). After applying multiple signal processing techniques to clean and strengthen the signal, an adaptive histogram-based thresholding method is used for apnea detection.

Summary. Tables 11 and 12 provide a summary of the work presented for sleep disorders detection. As can be seen from the tables, significant works have been conducted in sleep disorder detection especially in apnea detection. Wearable-based solutions have overall better accuracy. Both machine learning and signal processing are used to detect the disorders. Getting the ground truth for movement related disorders is one of the concerns in this category. Again, most of the experiments are simulating the sleep and apnea episodes rather than involving actual patients who are suffering from these disorders.

5 OPEN ISSUES AND FUTURE DIRECTIONS

Although significant work has been conducted in sleep monitoring using the unobtrusive and in-home solutions, there are still some open issues in this area that need to be addressed. In this section, we discuss some of those open issues and provide future research directions.

Table 11. Comparison of the Wearable-based Solutions for Disorders Detection

Work	Disorders	Sensor-Placement	Device Used	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
[50] 2017	PLMS	Accelerometer, gyroscope-foot dorsum	MetaMotionC	KNN	96.92%	Low	Low	Foot movements	13	Observation
AutoTag [97] 2018	Apnea	RFID tags-chest	Impinj R420, reader, Alien tags	LSTM, KL divergance	92%	Low	Low	Breathing from phase values of the reflected signal	4	NEULOG Respiration Belt Sensor
[22] 2018	OSA	Microphone-near nose	Self-made	Signal processing	97.00%	Medium	Medium	Acoustic signal from breathing	5	Manually
[71] 2019	Apnea	microphone-ear, pillow, bedside	Iphone4	Envelope detection, SVM	0.5 bpm (error), 80%	medium	Low	Acoustic signal	9	NEULOG Respiration Belt Sensor
[42] 2019	Apnea, hypopnea	Accelerometer, PPG-wrist, nasal airflow	Self-made	Signal processing, CMW	NA	Medium	Medium	SpO2, HR, RR	15	PSG
RestEaZe [8] 2020	RLS	Capacitive and inertial sensor-ankle	Self-made	SVM with linear kernel	90%	Low	Medium	Leg movements	12	PSG, IR camera

Table 12. Comparison of the Non-wearable-based Solutions for Disorders Detection

Work	Disorders	Sensor-Placement	Device Used	Contact	Methods Used	Accuracy	Obtrusiveness	Cost	Parameter Observed	# Subjects	Validation Method
[60] 2015	Central apnea, obstructive apnea, and hypopnea	Sonar-bedside	Four different smartphones	No	Signal processing, FMCW, peak detection	97%	Low	Low	Chest and abdomen motions	37	PSG + Video camera
[91] 2017	Apnea, hypopnea, PLM	Strain gauge sensors-bed	Mobility monitor by compliant concept	Yes	Fusion architecture, mRMR filter, NB	72%	Low	High	Pressure changes	9	PSG
EZ-Sleep [32] 2017	Insomnia	Radio-room	Self-made	No	Signal processing, FMCW, image processing, HMM, CNN	NA	Low	Low	Reflected signal	10	SleepProfiler
[86] 2018	Insomnia	Pressure sensors-bedsheet	Self-made	Yes	SVM with polynomial kernel	77.78%	Low	Low	Body pressure	9	AIS questionnaire
[75] 2020	Apnea	Fiber optic sensors-bedsheet	Self-made	Yes	Adaptive histogram-based thresholding	49.96%	Low	Low	Cardiac and chest movements	10	PSG

Trade-off between Accuracy and Comfort Level. Accuracy is always an important factor for any solutions. It is more important in areas like sleep monitoring, which consists of vital signs monitoring and sleep disorder detection. From the related literature, we have concluded that there is a tradeoff between the accuracy and the comfort level (unobtrusiveness) of the proposed solutions. Solutions that are more invasive in nature provide better accuracy for all four categories of sleep monitoring. However, solutions that are non-invasive and are easy to use, their accuracy is low in comparison especially in sleep stages classification and disorder detection. These in-home approaches can be used for general sleep monitoring but more work is required to bring the accuracy level to medical grade monitoring. There is a potential for further research to have solutions that are non-invasive and have the same level of accuracy as the clinical solutions.

Security and Privacy. In sleep monitoring, different sensors (wearable and non-wearable) collect the physiological data of the sleeping person. This data is then sent to either an edge device or to a central module (could be cloud) for further processing (recognition). In some cases, the processing entity is installed along with the sensors (in form of microcomputers, aurdinos, etc.) and the results are transferred via the network. From the related literature, we can see that the focus is mainly on the accuracy and ease of use. Very little (and mostly none) effort is made for the security and privacy of the collected data. Most of the data is medical data and should not be shared with unauthorized persons, but no specific measures are taken to protect the data from unauthorized access. Since the areas like telemedicine and remote monitoring are gaining popularity in healthcare, there is a need for adapting specific measures to protect the data. Hathaliya and Sudeep [29] studied the security and privacy issues in healthcare and identified some of the key challenges.

Limited Data. Data is a main aspect of sleep monitoring. Publicly available data sets are very important for evaluating and comparing the performance of any newly proposed solution. From the literature, we found that there are some data sets publicly available but most of them are of PSG data collected in a clinical environment. These data sets can help in the performance evaluation of solutions that use the same technique, i.e., PSG, ECG, and so on, but nowadays many solutions are unobtrusive and use wearable and now-wearable devices for monitoring physiological signals other than PSG. These solutions are implemented in home environments and use devices such as radio, RFID, accelerometer, magnetometer, pressure sensors, and WiFi. There are no data sets publicly available for these types of approaches except References [31, 68]. None of the related works covered in Section 4 except Reference [108] have released their data set. Since the use of intrusive techniques is increasing for sleep monitoring, there is a need for benchmark data sets to evaluate the performance of new solutions. The problem of limited data can be solved by user contributions, i.e., crowdsourcing the sleep data from their in-home devices. Also, the authors can be encouraged to share their data sets for validation purpose. Hence, large data sets can be created that are more balanced and diverse to reflect the real-world scenarios.

Need for Sleep-specific Devices. From the related literature, it can be seen that many devices that are used for sleep monitoring are the commercially available general purpose devices mainly used for activity tracking. Sleep monitoring is often considered as an additional feature of these devices. In some cases, the researchers have customized these devices to use it for sleep monitoring. Acceptance by the users is one of the main aspect of wearable devices. A study conducted by Shin et al. [81] reported that it is unreasonable to design an all-purpose wearable activity tracker. Although some sleep-specific devices exist in the market, there is a need of more research and development to produce specific sleep-monitoring devices that are cheap and easy to use for both short-term and long-term monitoring.

Use of Machine Learning. After collecting the physiological signals, two approaches can be used to infer the sleep related information. These approaches are signal processing and machine learning. We have found in the literature review that the use of machine learning is increasing for sleep monitoring. Many studies have used machine learning and have explored the feature engineering to get good results. One of the advantages of using machine learning is the high accuracy for sleep monitoring especially for detecting disorders, sleep stages, and postures. However, machine learning requires more resources, which is a problem for wearable devices given their small sizes and battery requirements. Some solutions have tried to tackle this problem by shifting the training phase offline [40] while other have used light weight techniques to ease the burden on the wearable devices.

Long-term Versus Short-term Monitoring. The related literature shows that unobtrusive approaches (both wearable and non-wearable) are evaluated only for the short-term monitoring (mostly a single night or a few hours). However, it would be interesting to see the performance for a long-term monitoring. Despite the low accuracy of unobtrusive approaches as compared to clinical solutions, an advantage of the unobtrusive solutions is their ease of use and low cost for long-term deployment. These solutions can be explored for long-term in-home monitoring and can help in the early diagnosis of many health relate problems such as epilepsy [55], depression [21], and other respiratory problems [5].

6 CONCLUSION

Sleep monitoring has been an important approach to understanding sleep behaviors and improving sleep quality. With the recent advancements in sensor technologies, unobtrusive sleep monitoring using wearable or non-wearable sensors has become an active research area. In this article, we have presented a comprehensive review of the latest research efforts in the area of sleep monitoring. We divide the related literature into four categories: sleep stages classification, sleep postures recognition, sleep disorders detection, and vital signs monitoring. Unlike the previous surveys, which focus on some of the categories in sleep monitoring, we cover all four categories. The main focus of this survey is the unobtrusive techniques used for in-home sleep monitoring. We review the latest work in all four categories of sleep monitoring and provide a comparison based on 10 key factors. We also identify and discuss some open research issues and provide future research directions to stimulate further research in this important topic.

REFERENCES

[1] Khald Ali I. Aboalayon, Miad Faezipour, Wafaa S. Almuhammadi, and Saeid Moslehpour. 2016. Sleep stage classification using EEG signal analysis: A comprehensive survey and new investigation. *Entropy* 18, 9 (2016), 272.

[2] Toni Adame, Albert Bel, Anna Carreras, Joan Melia-Segui, Miquel Oliver, and Rafael Pous. 2018. CUIDATS: An RFID–WSN hybrid monitoring system for smart healthcare environments. *Future Gen. Comput. Syst.* 78 (2018), 602–615.

[3] Sahar Ahmadzadeh, Jikui Luo, and Richard Wiffen. 2020. Review on biomedical sensors, technologies and algorithms for diagnosis of sleep disordered breathing: Comprehensive survey. *IEEE Rev. Biomed. Engineer.* (2020). DOI: 10.1109/RBME.2020.3033930

[4] Mostafa Alizadeh, George Shaker, João Carlos Martins De Almeida, Plinio Pelegrini Morita, and Safeddin Safavi-Naeini. 2019. Remote monitoring of human vital signs using mm-wave FMCW radar. *IEEE Access* 7 (2019), 54958–54968.

[5] Patrick Bachour, Adel Bachour, Paula Kauppi, Paula Maasilta, Antti Mäkitie, and Tuula Palotie. 2016. Oral appliance in sleep apnea treatment: Respiratory and clinical effects and long-term adherence. *Sleep Breath.* 20, 2 (2016), 805–812.

[6] Paolo Barsocchi, Monica Bianchini, Antonino Crivello, Davide La Rosa, Filippo Palumbo, and Franco Scarselli. 2016. An unobtrusive sleep monitoring system for the human sleep behaviour understanding. In *Proceedings of the 7th IEEE International Conference on Cognitive Infocommunications (CogInfoCom'16)*. IEEE, 000091–000096.

[7] Zachary Beattie, Y. Oyang, A. Statan, A. Ghoreyshi, A. Pantelopoulos, A. Russell, and C. J. P. M. Heneghan. 2017. Estimation of sleep stages in a healthy adult population from optical plethysmography and accelerometer signals. *Physiol. Measure.* 38, 11 (2017), 1968.

[8] Stanislav Bobovych, Fahad Sayeed, Nilanjan Banerjee, Ryan Robucci, and Richard P. Allen. 2020. RestEaZe: Low-power accurate sleep monitoring using a wearable multi-sensor ankle band. *Smart Health* 16 (2020), 100113.

[9] Alexander J. Boe, Lori L. McGee Koch, Megan K. O'Brien, Nicholas Shawen, John A. Rogers, Richard L. Lieber, Kathryn J. Reid, Phyllis C. Zee, and Arun Jayaraman. 2019. Automating sleep stage classification using wireless, wearable sensors. *NPJ Dig. Med.* 2, 1 (2019), 1–9.

[10] Reza Boostani, Foroozan Karimzadeh, and Mohammad Nami. 2017. A comparative review on sleep stage classification methods in patients and healthy individuals. *Comput. Methods Programs Biomed.* 140 (2017), 77–91.

[11] Marko Borazio, Eugen Berlin, Nagihan Kücükyildiz, Philipp Scholl, and Kristof Van Laerhoven. 2014. Towards benchmarked sleep detection with wrist-worn sensing units. In *Proceedings of the IEEE International Conference on Healthcare Informatics*. IEEE, 125–134.

[12] Dina Brooks, Richard L. Horner, Louise F. Kozar, Caroline L. Render-Teixeira, Eliot A. Phillipson, et al. 1997. Obstructive sleep apnea as a cause of systemic hypertension. Evidence from a canine model. *J. Clin. Investig.* 99, 1 (1997), 106–109.

[13] Liqiong Chang, Jiaqi Lu, Ju Wang, Xiaojiang Chen, Dingyi Fang, Zhanyong Tang, Petteri Nurmi, and Zheng Wang. 2018. SleepGuard: Capturing rich sleep information using smartwatch sensing data. *Proc. ACM Interact., Mobile, Wear. Ubiq. Technol.* 2, 3 (2018), 1–34.

[14] Xiangmao Chang, Cheng Peng, Guoliang Xing, Tian Hao, and Gang Zhou. 2020. iSleep: A smartphone system for unobtrusive sleep quality monitoring. *ACM Trans. Sensor Netw.* 16, 3 (2020), 1–32.

[15] Zhenyu Chen, Mu Lin, Fanglin Chen, Nicholas D. Lane, Giuseppe Cardone, Rui Wang, Tianxing Li, Yiqiang Chen, Tanzeem Choudhury, and Andrew T. Campbell. 2013. Unobtrusive sleep monitoring using smartphones. In *Proceedings of the 7th International Conference on Pervasive Computing Technologies for Healthcare and Workshops*. IEEE, 145–152.

[16] Michael Chu, Thao Nguyen, Vaibhav Pandey, Yongxiao Zhou, Hoang N. Pham, Ronen Bar-Yoseph, Shlomit Radom-Aizik, Ramesh Jain, Dan M. Cooper, and Michelle Khine. 2019. Respiration rate and volume measurements using wearable strain sensors. *NPJ Dig. Med.* 2, 1 (2019), 1–9.

[17] J. Clemente, M. Valero, F. Li, C. Wang, and W. Song. 2020. Helena: Real-time contact-free monitoring of sleep activities and events around the bed. In *Proceedings of the IEEE International Conference on Pervasive Computing and Communications (PerCom'20)*. 1–10. https://doi.org/10.1109/PerCom45495.2020.9127369

[18] Joseph De Koninck, Pierre Gagnon, and Serge Lallier. 1983. Sleep positions in the young adult and their relationship with the subjective quality of sleep. *Sleep* 6, 1 (1983), 52–59.

[19] Sarah Fallmann and Liming Chen. 2019. Computational sleep behavior analysis: A survey. *IEEE Access* 7 (2019), 142421–142440.

[20] Sarah Fallmann, Rick van Veen, Liming Chen, David Walker, Feng Chen, and Chengsheng Pan. 2017. Wearable accelerometer-based extended sleep position recognition. In *Proceedings of the IEEE 19th International Conference on e-Health Networking, Applications and Services (Healthcom'17)*. IEEE, 1–6.

[21] Beifang Fan, Tian Wang, Wanxin Wang, Sheng Zhang, Meiqian Gong, Wenyan Li, Ciyong Lu, and Lan Guo. 2019. Long-term exposure to ambient fine particulate pollution, sleep disturbance and their interaction effects on suicide attempts among chinese adolescents. *J. Affect. Disorders* 258 (2019), 89–95.

[22] Yu Fang, Zhongwei Jiang, and Haibin Wang. 2018. A novel sleep respiratory rate detection method for obstructive sleep apnea based on characteristic moment waveform. *J. Healthcare Eng.* (2018).

[23] Rayan Gargees, James M. Keller, Mihail Popescu, and Marjorie Skubic. 2019. Non-invasive classification of sleep stages with a hydraulic bed sensor using deep learning. In *Proceedings of the International Conference on Smart Homes and Health Telematics.* Springer, 73–82.

[24] M. M. Ghassemi, B. E. Moody, L. H. Lehman, C. Song, Q. Li, H. Sun, R. G. Mark, M. B. Westover, and G. D. Clifford. 2018. You snooze, you win: the PhysioNet/computing in cardiology challenge 2018. In *Proceedings of the Computing in Cardiology Conference (CinC'18)*, Vol. 45. 1–4. https://doi.org/10.22489/CinC.2018.049

[25] Laurent Giovangrandi, Omer T. Inan, Richard M. Wiard, Mozziyar Etemadi, and Gregory T. A. Kovacs. 2011. Ballistocardiography'a method worth revisiting. In *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society.* IEEE, 4279–4282.

[26] Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. 2000. PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation* 101, 23 (2000), e215–e220.

[27] Weixi Gu, Longfei Shangguan, Zheng Yang, and Yunhao Liu. 2015. Sleep hunter: Towards fine grained sleep stage tracking with smartphones. *IEEE Trans. Mobile Comput.* 15, 6 (2015), 1514–1527.

[28] Tian Hao, Guoliang Xing, and Gang Zhou. 2013. isleep: Unobtrusive sleep quality monitoring using smartphones. In *Proceedings of the 11th ACM Conference on Embedded Networked Sensor Systems.* 1–14.

[29] Jigna J. Hathaliya and Sudeep Tanwar. 2020. An exhaustive survey on security and privacy issues in healthcare 4.0. *Comput. Commun.* 153 (2020), 311–335.

[30] Conor Heneghan. 2011. St. Vincent's University Hospital/University College Dublin Sleep Apnea Database. DOI: 10.13026/C26C7D

[31] Peter Hillyard, Anh Luong, Alemayehu Solomon Abrar, Neal Patwari, Krishna Sundar, Robert Farney, Jason Burch, Christina Porucznik, and Sarah Hatch Pollard. 2018. Experience: Cross-technology radio respiratory monitoring performance study. In *Proceedings of the 24th Annual International Conference on Mobile Computing and Networking.* 487–496.

[32] Chen-Yu Hsu, Aayush Ahuja, Shichao Yue, Rumen Hristov, Zachary Kabelac, and Dina Katabi. 2017. Zero-effort in-home sleep and insomnia monitoring using radio signals. *Proc. ACM Interact., Mobile, Wear. Ubiq. Technol.* 1, 3 (2017), 1–18.

[33] Xiaoxuan Hu, Kagome Naya, Peng Li, Toshiaki Miyazaki, Kun Wang, and Yanfei Sun. 2018. Non-invasive sleeping posture recognition and body movement detection based on RFID. In *Proceedings of the IEEE International Conference on Internet of Things (iThings'18) and IEEE Green Computing and Communications (GreenCom'18) and IEEE Cyber, Physical and Social Computing (CPSCom'18) and IEEE Smart Data (SmartData'18).* IEEE, 1817–1820.

[34] Zawar Hussain, Subhash Sagar, Wei Emma Zhang, and Quan Z. Sheng. 2019. A cost-effective and non-invasive system for sleep and vital signs monitoring using passive RFID tags. In *Proceedings of the 16th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services.* 153–161.

[35] Zawar Hussain, Quan Z. Sheng, and Wei Emma Zhang. 2020. A review and categorization of techniques on device-free human activity recognition. *J. Netw. Comput. Appl.* 167 (Oct. 2020), 102738.

[36] Vanessa Ibáñez, Josep Silva, and Omar Cauli. 2018. A survey on sleep questionnaires and diaries. *Sleep Med.* 42 (2018), 90–96.

[37] Conrad Iber, Sonia Ancoli-Israel, Andrew L. Chesson, Stuart F. Quan, et al. 2007. *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications.* Vol. 1. American Academy of Sleep Medicine, Westchester, IL.

[38] O. T. Inan, P. F. Migeotte, K. S. Park, M. Etemadi, K. Tavakolian, R. Casanella, J. Zanetti, J. Tank, I. Funtova, G. K. Prisk, et al. 2015. Ballistocardiography and seismocardiography: A review of recent advances. *IEEE J. Biomed. Health Inform.* 19, 4 (2015), 1414–1427.

[39] Yoshiki Ishibashi and Akiyoshi Shimura. 2020. Association between work productivity and sleep health: A cross-sectional study in Japan. *Sleep Health* 6, 3 (2020), 270–276.

[40] Po-Yuan Jeng, Li-Chun Wang, Chaur-Jong Hu, and Dean Wu. 2021. A wrist sensor sleep posture monitoring system: An automatic labeling approach. *Sensors* 21, 1 (2021), 258.

[41] Sanghoon Jeon, Taejoon Park, Anand Paul, Yang-Soo Lee, and Sang Hyuk Son. 2019. A wearable sleep position tracking system based on dynamic state transition framework. *IEEE Access* 7 (2019), 135742–135756.

[42] Yeong Jun Jeon and Soon Ju Kang. 2019. Wearable sleepcare kit: Analysis and prevention of sleep apnea symptoms in real-time. *IEEE Access* 7 (2019), 60634–60649.

[43] Zhenhua Jia, Amelie Bonde, Sugang Li, Chenren Xu, Jingxian Wang, Yanyong Zhang, Richard E. Howard, and Pei Zhang. 2017. Monitoring a person's heart rate and respiratory rate on a shared bed using geophones. In *Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems.* 1–14.

[44] Juhani Kalsi, Timo Tervo, Adel Bachour, and Markku Partinen. 2018. Sleep versus non-sleep-related fatal road accidents. *Sleep Med.* 51 (2018), 148–152.

[45] Jessica M. Kelly, Robert E. Strecker, and Matt T. Bianchi. 2012. Recent developments in home sleep-monitoring devices. *ISRN Neurol.* 2012.
[46] Sirvan Khalighi, Teresa Sousa, José Moutinho Santos, and Urbano Nunes. 2016. ISRUC-sleep: A comprehensive public dataset for sleep researchers. *Comput. Methods Prog. Biomed.* 124 (2016), 180–192.
[47] Ali Kiaghadi, Seyedeh Zohreh Homayounfar, Jeremy Gummeson, Trisha Andrew, and Deepak Ganesan. 2019. Phyjama: Physiological sensing via fiber-enhanced pyjamas. *Proc. ACM Interact., Mobile, Wear. Ubiq. Technol.* 3, 3 (2019), 1–29.
[48] Ali Kiaghadi, S. Zohreh Homayounfar, Jeremy Gummeson, Trisha L. Andrew, and Deepak Ganesan. 2020. Phyjama: Physiological sensing via fiber-enhanced pyjamas. *GetMobile: Mobile Comput. Commun.* 24, 1 (2020), 33–37.
[49] Chang-Sei Kim, Stephanie L. Ober, M. Sean McMurtry, Barry A. Finegan, Omer T. Inan, Ramakrishna Mukkamala, and Jin-Oh Hahn. 2016. Ballistocardiogram: Mechanism and potential for unobtrusive cardiovascular health monitoring. *Sci. Rep.* 6 (2016), 31297.
[50] Saewon Kye, Junhyung Moon, Taeho Lee, Sangyeop Lee, Kyoungwoo Lee, Seung-chul Shin, and Yong Seung Lee. 2017. Detecting periodic limb movements in sleep using motion sensor embedded wearable band. In *Proceedings of the IEEE International Conference on Systems, Man, and Cybernetics (SMC'17).* IEEE, 1087–1092.
[51] F. Lin, Y. Zhuang, C. Song, A. Wang, Y. Li, C. Gu, C. Li, and W. Xu. 2017. SleepSense: A noncontact and cost-effective sleep monitoring system. *IEEE Trans. Biomed. Circ. Syst.* 11, 1 (2017), 189.
[52] Jia Liu, Xingyu Chen, Shigang Chen, Xiulong Liu, Yanyan Wang, and Lijun Chen. 2019. Tagsheet: Sleeping posture recognition with an unobtrusive passive tag matrix. In *Proceedings of the IEEE Conference on Computer Communications (INFOCOM'19).* IEEE, 874–882.
[53] Jian Liu, Yingying Chen, Yan Wang, Xu Chen, Jerry Cheng, and Jie Yang. 2018. Monitoring vital signs and postures during sleep using WiFi signals. *IEEE Internet Things J.* 5, 3 (2018), 2071–2084.
[54] Jianchao Lu, Xi Zheng, Quan Z. Sheng, Zawar Hussain, Jiaxing Wang, and Wanlei Zhou. 2019. Mfe-har: Multiscale feature engineering for human activity recognition using wearable sensors. In *Proceedings of the 16th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services.* 180–189.
[55] Brendan P. Lucey, Averi Leahy, Regine Rosas, and Paul J. Shaw. 2015. A new model to study sleep deprivation-induced seizure. *Sleep* 38, 5 (2015), 777–785.
[56] Sam Mansfield, Katia Obraczka, and Shuvo Roy. 2019. Pressure injury prevention: A survey. *IEEE Rev. Biomed. Eng.* 13 (2019), 352–368.
[57] Georges Matar, Jean-Marc Lina, Julie Carrier, and Georges Kaddoum. 2018. Unobtrusive sleep monitoring using cardiac, breathing and movements activities: An exhaustive review. *IEEE Access* 6 (2018), 45129–45152.
[58] Stefano Milici, Antonio Lázaro, Ramon Villarino, David Girbau, and Marco Magnarosa. 2018. Wireless wearable magnetometer-based sensor for sleep quality monitoring. *IEEE Sensors J.* 18, 5 (2018), 2145–2152.
[59] T. Nakamura, Y. D. Alqurashi, M. J. Morrell, and D. P. Mandic. 2020. Hearables: Automatic overnight sleep monitoring with standardized in-ear EEG sensor. *IEEE Trans. Biomed. Eng.* 67, 1 (2020), 203.
[60] Rajalakshmi Nandakumar, Shyamnath Gollakota, and Nathaniel Watson. 2015. Contactless sleep apnea detection on smartphones. In *Proceedings of the 13th Annual International Conference on Mobile Systems, Applications, and Services.* 45–57.
[61] Christian O'Reilly, Nadia Gosselin, Julie Carrier, and Tore Nielsen. 2014. Montreal archive of sleep studies: An open-access resource for instrument benchmarking and exploratory research. *J. Sleep Res.* 23, 6 (2014), 628–635.
[62] Qiang Pan, Damien Brulin, and Eric Campo. 2020. Current status and future challenges of sleep monitoring systems: Systematic review. *JMIR Biomed. Eng.* 5, 1 (2020), e20921.
[63] Lori A. Panossian and Alon Y. Avidan. 2009. Review of sleep disorders. *Med. Clinics North Amer.* 93, 2 (2009), 407–425.
[64] Kwang Suk Park and Sang Ho Choi. 2019. Smart technologies toward sleep monitoring at home. *Biomed. Eng. Lett.* 9, 1 (2019), 73–85.
[65] Jonathan D. Pegan, Jasmine Zhang, Michael Chu, Thao Nguyen, Sun-Jun Park, Akshay Paul, Joshua Kim, Mark Bachman, and Michelle Khine. 2016. Skin-mountable stretch sensor for wearable health monitoring. *Nanoscale* 8, 39 (2016), 17295–17303.
[66] Thomas Penzel, George B. Moody, Roger G. Mark, Ary L. Goldberger, and J. Hermann Peter. 2000. The apnea-ECG database. In *Computers in Cardiology 2000, Vol. 27 (Cat. 00CH37163).* IEEE, 255–258.
[67] Ignacio Perez-Pozuelo, Bing Zhai, Joao Palotti, Raghvendra Mall, Michaël Aupetit, Juan M. Garcia-Gomez, Shahrad Taheri, Yu Guan, and Luis Fernandez-Luque. 2020. The future of sleep health: A data-driven revolution in sleep science and medicine. *NPJ Dig. Med.* 3, 1 (2020), 1–15.
[68] M. Baran Pouyan, Javad Birjandtalab, Mehrdad Heydarzadeh, Mehrdad Nourani, and Sarah Ostadabbas. 2017. A pressure map dataset for posture and subject analytics. In *Proceedings of the IEEE EMBS International Conference on Biomedical & Health Informatics (BHI'17).* IEEE, 65–68.
[69] Stuart F. Quan, Barbara V. Howard, Conrad Iber, James P. Kiley, F. Javier Nieto, George T. O'Connor, David M. Rapoport, Susan Redline, John Robbins, Jonathan M. Samet, et al. 1997. The sleep heart health study: Design, rationale, and methods. *Sleep* 20, 12 (1997), 1077–1085.
[70] A. Rechtschaffen and A. Kales. 1968. *A Manual of Standardized Terminology, Techniques and Scoring System for Sleep Stages of Human Subjects: NIH Publication No. 204.* Public Health Service, National Institutes of Health, Bethesda, MD.
[71] Yanzhi Ren, Chen Wang, Yingying Chen, Jie Yang, and Hongwei Li. 2019. Noninvasive fine-grained sleep monitoring leveraging smartphones. *IEEE Internet Things J.* 6, 5 (2019), 8248–8261.

[72] Yanzhi Ren, Chen Wang, Jie Yang, and Yingying Chen. 2015. Fine-grained sleep monitoring: Hearing your breathing with smartphones. In *Proceedings of the IEEE Conference on Computer Communications (INFOCOM'15)*. IEEE, 1194–1202.

[73] Scott Rickard. 2007. The DUET blind source separation algorithm. In *Blind Speech Separation*. Springer, 217–241.

[74] Ibrahim Sadek, Antoine Demarasse, and Mounir Mokhtari. 2019. Internet of things for sleep tracking: Wearables vs. nonwearables. *Health Technol.* 10 (2019), 333–340.

[75] Ibrahim Sadek, Terry Tan Soon Heng, Edwin Seet, and Bessam Abdulrazak. 2020. A new approach for detecting sleep apnea using a contactless bed sensor: Comparison study. *J. Med. Internet Res.* 22, 9 (2020), e18297.

[76] Ibrahim Sadek and Mounir Mohktari. 2018. Nonintrusive remote monitoring of sleep in home-based situation. *J. Med. Syst.* 42, 4 (2018), 64.

[77] Ibrahim Sadek, Edwin Seet, Jit Biswas, Bessam Abdulrazak, and Mounir Mokhtari. 2017. Nonintrusive vital signs monitoring for sleep apnea patients: A preliminary study. *IEEE Access* 6 (2017), 2506–2514.

[78] Kristin E. Schwab, Bonnie Ronish, Dale M. Needham, An Q. To, Jennifer L. Martin, and Biren B. Kamdar. 2018. Actigraphy to evaluate sleep in the intensive care unit. A systematic review. *Ann. Amer. Thoracic Soc.* 15, 9 (2018), 1075–1082.

[79] Chamara V. Senaratna, Jennifer L. Perret, Caroline J. Lodge, Adrian J. Lowe, Brittany E. Campbell, Melanie C. Matheson, Garun S. Hamilton, and Shyamali C. Dharmage. 2017. Prevalence of obstructive sleep apnea in the general population: A systematic review. *Sleep Med. Rev.* 34 (2017), 70–81.

[80] Pragya Sharma and Edwin C. Kan. 2018. Sleep scoring with a UHF RFID tag by near field coherent sensing. In *Proceedings of the IEEE/MTT-S International Microwave Symposium-IMS*. IEEE, 1419–1422.

[81] Grace Shin, Mohammad Hossein Jarrahi, Yu Fei, Amir Karami, Nicci Gafinowitz, Ahjung Byun, and Xiaopeng Lu. 2019. Wearable activity trackers, accuracy, adoption, acceptance and health impact: A systematic literature review. *J. Biomed. Inform.* 93 (2019), 103153.

[82] Michael T. Smith, Christina S. McCrae, Joseph Cheung, Jennifer L. Martin, Christopher G. Harrod, Jonathan L. Heald, and Kelly A. Carden. 2018. Use of actigraphy for the evaluation of sleep disorders and circadian rhythm sleep-wake disorders: An American academy of sleep medicine systematic review, meta-analysis, and GRADE assessment. *J. Clin. Sleep Med.* 14, 7 (2018), 1209–1230.

[83] Karine Spiegel, Kristen Knutson, Rachel Leproult, Esra Tasali, and Eve Van Cauter. 2005. Sleep loss: A novel risk factor for insulin resistance and type 2 diabetes. *J. Appl. Physiol.* 99, 5 (2005), 2008–2019.

[84] Xiao Sun, Li Qiu, Yibo Wu, Yeming Tang, and Guohong Cao. 2017. Sleepmonitor: Monitoring respiratory rate and body position during sleep using smartwatch. *Proc. ACM Interact. Mobile, Wear. Ubiq. Technol.* 1, 3 (2017), 1–22.

[85] Nico Surantha, Gede Putra Kusuma, and Sani M. Isa. 2016. Internet of things for sleep quality monitoring system: A survey. In *Proceedings of the 11th International Conference on Knowledge, Information and Creativity Support Systems (KICSS)*. IEEE, 1–6.

[86] Suthipat Swangarom, Takuya Tajima, Takehiko Abe, and Haruhiko Kimura. 2018. A proposal for a sleep disorder detection system. *Sensors Mater.* 30, 7 (2018), 1437–1446.

[87] Alexander Tataraidze, Lyudmila Korostovtseva, Lesya Anishchenko, Mikhail Bochkarev, Yurii Sviryaev, and Sergey Ivashov. 2016. Bioradiolocation-based sleep stage classification. In *Proceedings of the 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'16)*. IEEE, 2839–2842.

[88] Vinh Phuc Tran, Adel Ali Al-Jumaily, and Syed Mohammed Shamsul Islam. 2019. Doppler radar-based non-contact health monitoring for obstructive sleep apnea diagnosis: A comprehensive review. *Big Data Cogn. Comput.* 3, 1 (2019), 3.

[89] Emmi Turppa, Juha M. Kortelainen, Oleg Antropov, and Tero Kiuru. 2020. Vital sign monitoring using FMCW radar in various sleeping scenarios. *Sensors* 20, 22 (2020), 6505.

[90] Robert D. Vorona, Maria P. Winn, Teresa W. Babineau, Benjamin P. Eng, Howard R. Feldman, and J. Catesby Ware. 2005. Overweight and obese patients in a primary care population report less sleep than patients with a normal body mass index. *Arch. Internal Med.* 165, 1 (2005), 25–30.

[91] D. Waltisberg, O. Amft, D. P. Brunner, and G. Troster. 2017. Detecting disordered breathing and limb movement using in-bed force sensors. *IEEE J. Biomed. Health Inform.* 21, 4 (2017), 930.

[92] Chuyu Wang, Lei Xie, Wei Wang, Yingying Chen, Yanling Bu, and Sanglu Lu. 2018. Rf-ecg: Heart rate variability assessment based on cots rfid tag array. *Proc. ACM Interact. Mobile, Wear. Ubiq. Technol.* 2, 2 (2018), 1–26.

[93] Yanwen Wang and Yuanqing Zheng. 2020. TagBreathe: Monitor breathing with commodity RFID systems. *IEEE Trans. Mobile Comput.* 19, 4 (2020), 969–981.

[94] Hau-tieng Wu, Ronen Talmon, and Yu-Lun Lo. 2014. Assess sleep stage by modern signal processing techniques. *IEEE Trans. Biomed. Eng.* 62, 4 (2014), 1159–1168.

[95] Zizhen Xie, Fei Chen, William A. Li, Xiaokun Geng, Changhong Li, Xiaomei Meng, Yan Feng, Wei Liu, and Fengchun Yu. 2017. A review of sleep disorders and melatonin. *Neurol. Res.* 39, 6 (2017), 559–565.

[96] Xiaowei Xu, Feng Lin, Aosen Wang, Yu Hu, Ming-Chun Huang, and Wenyao Xu. 2016. Body-earth mover's distance: A matching-based approach for sleep posture recognition. *IEEE Trans. Biomed. Circ. Syst.* 10, 5 (2016), 1023–1035.

[97] Chao Yang, Xuyu Wang, and Shiwen Mao. 2018. AutoTag: Recurrent variational autoencoder for unsupervised apnea detection with RFID tags. In *Proceedings of the IEEE Global Communications Conference (GLOBECOM'18)*. IEEE, 1–7.

[98] Yanni Yang, Jiannong Cao, Xuefeng Liu, and Kai Xing. 2018. Multi-person sleeping respiration monitoring with COTS WiFi devices. In *Proceedings of the IEEE 15th International Conference on Mobile Ad Hoc and Sensor Systems (MASS'18)*. IEEE, 37–45.
[99] Zhicheng Yang, Parth H. Pathak, Yunze Zeng, Xixi Liran, and Prasant Mohapatra. 2017. Vital sign and sleep monitoring using millimeter wave. *ACM Trans. Sensor Netw.* 13, 2 (2017), 1–32.
[100] Ruhan Yi, Moein Enayati, James M. Keller, Mihail Popescu, and Marjorie Skubic. 2019. Non-invasive in-home sleep stage classification using a ballistocardiography bed sensor. In *Proceedings of the IEEE EMBS International Conference on Biomedical and Health Informatics (BHI'19)*. IEEE, 1–4.
[101] Ozgur Yilmaz and Scott Rickard. 2004. Blind separation of speech mixtures via time-frequency masking. *IEEE Trans. Signal Process.* 52, 7 (2004), 1830–1847.
[102] Shichao Yue, Hao He, Hao Wang, Hariharan Rahul, and Dina Katabi. 2018. Extracting multi-person respiration from entangled rf signals. *Proc. ACM Interact. Mobile, Wear. Ubiq. Technol.* 2, 2 (2018), 1–22.
[103] Shichao Yue, Yuzhe Yang, Hao Wang, Hariharan Rahul, and Dina Katabi. 2020. BodyCompass: Monitoring sleep posture with wireless signals. *Proc. ACM Interact. Mobile, Wear. Ubiq. Technol.* 4, 2 (2020), 1–25.
[104] F. Zhang, C. Wu, B. Wang, M. Wu, D. Bugos, H. Zhang, and K. J. R. Liu. 2021. SMARS: Sleep monitoring via ambient radio signals. *IEEE Trans. Mobile Comput.* 20, 1 (2021), 217–231. https://doi.org/10.1109/TMC.2019.2939791
[105] Li Zhang, Junjun Xiong, Heng Zhao, Hong Hong, Xiaohua Zhu, and Changzhi Li. 2017. Sleep stages classification by CW doppler radar using bagged trees algorithm. In *Proceedings of the IEEE Radar Conference (RadarConf'17)*. IEEE, 0788–0791.
[106] Xin Zhang, Weixuan Kou, I. Eric, Chao Chang, He Gao, Yubo Fan, and Yan Xu. 2018. Sleep stage classification based on multi-level feature learning and recurrent neural networks via wearable device. *Comput. Biol. Med.* 103 (2018), 71–81.
[107] Yuezhou Zhang, Zhicheng Yang, Ke Lan, Xiaoli Liu, Zhengbo Zhang, Peiyao Li, Desen Cao, Jiewen Zheng, and Jianli Pan. 2019. Sleep stage classification using bidirectional lstm in wearable multi-sensor systems. In *Proceedings of the IEEE INFOCOM Conference on Computer Communications Workshops (INFOCOM WKSHPS'19)*. IEEE, 443–448.
[108] Mingmin Zhao, Shichao Yue, Dina Katabi, Tommi S. Jaakkola, and Matt T. Bianchi. 2017. Learning sleep stages from radio signals: A conditional adversarial architecture. In *Proceedings of the International Conference on Machine Learning*. 4100–4109.

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